

Joint Adaptive Target Detection, Position Determination, and Velocity Estimation With a Long-Baseline Radar Network

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This article investigates the problem of joint adaptive target detection, position determination, and velocity estimation with a long-baseline radar network. The problem is cast as a binary composite hypothesis testing in a homogeneous noise environment, where the target's position and velocity under the alternative hypothesis are unknown. To tackle the problem, the generalized likelihood ratio testing criterion (both one-step and two-step) is considered, where the statistics of the maximum likelihood estimates (MLEs) of the parameters are calculated. The optimization problem involved by the maximum likelihood estimation is handled via two different optimization strategies: the direct refinement method based on the steepest ascent initialized by the given rough point and the mixed refinement initialized by indirect results. In this way, the MLEs of the position and velocity are directly provided, while the target is detected with less straddle loss. The detection performance and parameter estimation

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accuracy of the proposed adaptive architectures are analyzed. Furthermore, the simulation results are compared with the “detection then localization” methods to illustrate the validity of the proposed method.

I. INTRODUCTION

Radar networks where signals from all separated stations are fused and jointly processed provide several advantages compared with conventional monostatic radars [1]. The multiradar system can significantly improve target detection sensitivity [2], [3], [4], localization accuracy [5], [6], [7], and interference proofing ability [8], [9], [10] with suitable configurations and data fusion algorithms.

In practical work, radar network systems can be roughly divided into two classes according to base lengths (distances between stations): short- and long-baseline radar networks [11]. The former contains several radar nodes where effective base lengths are essentially less than expected target ranges in the main part of the coverage area. In this configuration, geometrical relationships between signal parameters and target coordinates, as well as most processing algorithms, are similar to monostatic radar. With a coherent or noncoherent combination, they can extend the coverage area [12], [13]. Increasing the network baseline is often desirable to capture diversity gain and geometry gain. If effective base lengths are of the same order or greater than expected target ranges (in the main part of the coverage area), such systems may be considered as radar networks with long baseline distances. Radar networks with long baseline distances can provide higher performance in position determination [14], [15], [16] and velocity measurement [17].

Historically, the detection and localization are treated as two isolated subproblems in the multiradar system, that is, detection then localization [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44]. Two frameworks have been testified based on the idea of detection and then localization by many references in the open literature, i.e., the distributed and the centralized. In the distributed framework, global decision and localization are activated after local decision and estimation events are triggered. The measurements, which include time of arrival (TOA), frequency of arrival, or angle of arrival, are first extracted from each local radar station. Then, the final decision is made via a distributed fusion rule, such as least squares technique [18], [19], [20], [21], weighted decision fusion [22], [23], likelihood ratio test (LRT) [24], maximum likelihood (ML) estimation [25], or optimization algorithms [26]. There are also some methods suitable for asynchronous multistatic radar systems [27], [28], [29]. If the presence of a target is declared, the target's location can be obtained by solving the geometric equations based on the local measurements [30], [31], [32], which are known as the indirect localization method. However, because the radar cross section and path attenuation vary in different radars, large estimated errors may occur in some local radar

nodes, which causes a degradation of final detection and localization performance.

To handle the issue, centralized methods are studied, where the raw signals received by all radars are transmitted to the fusion center for joint processing [33], [34]. This framework takes full advantage of the global nondestructive information compared with the distributed approach [33], [35], thereby enhancing detection and localization performance, particularly in scenarios with a low signal-to-noise ratio. The generalized likelihood ratio test (GLRT), the Rao test, and the Wald test are three main multichannel adaptive detection criteria [36], [37], [38]. They jointly use the data of cells under test and secondary data to construct the detector for adaptive detection. The adaptive detection not only achieves the function of filtering and constant false alarm rate (CFAR) processing simultaneously, but also has simple detection procedure [39]. In this case, the detection problem is formulated as a binary hypothesis testing problem. This article solves this problem through the GLRT criterion [40], [41], [42]. The location is directly extracted from raw signals if the target is declared [43], [44]. However, the target signal's peak will decrease when its position falls between samples, which is called straddle loss, which causes a large degradation in detection performance.

Thus, joint target detection and parameter estimation are pursued, providing the target parameters simultaneously with the detection operation. As discussed in [45], detection and estimation problems can be handled in the same manner, and solving these two subproblems with one step shows a higher performance than "detection then estimation." The concept of joint target detection and estimation has also been explored in other studies within various localization domains, including range [46], motion parameters [47], and angle [48]. However, joint target detection and localization has only garnered limited attention concerning the multi-radar system. Specifically, study [49] considers the joint detection and localization within the context of the Gaussian white noise with a known variance. In [50], the algorithm of simultaneous processing is also developed for distributed multiple-input multiple-output radars. The method can detect and position multiple targets with a multiradar system but is only suitable for the stationary target in the 2-D coordinate system.

The simultaneous adaptive moving target detection, position determination, and velocity estimation has not yet been studied using a long-baseline radar network in the more general 3-D scenario. Following the idea of joint detection and localization [49], [50], we investigate an adaptive framework within the context of Gaussian interference with unknown spectral properties. First, the detection problem is formulated as a composite hypothesis testing problem, where the clutter-plus-noise covariance matrix and some target parameters (i.e., target position, velocity, and echo amplitude) are modeled as unknown parameters. Then, the adaptive architectures are devised based on the GLRT criterion, including both one-step GLRT [51], [52] and adaptive matched filter (AMF) [53], [54], where the

secondary data free of the target echo is exploited to obtain ML estimation of the noise-plus-clutter parameters. The target's position and velocity are solved by the further optimization problem. Two optimization strategies are proposed when a rough solution is given: a direct refinement strategy based on the steepest ascent method and a mixed refinement strategy where the direct refinement is initialized with the output of the indirect method. In this way, the proposed architecture can detect targets with less straddle loss and, at the same time, determine the accurate position and velocity estimates directly. Finally, the detection performance of the adaptive framework proposed in this article is compared with that of the distributed and mismatched centralized detector. In addition, the Cramer–Rao lower bound (CRLB) of the target position and velocity estimates is calculated for accuracy analysis. The root-mean-square error (RMSE) of the estimates is compared with the indirect method. The experimental results corroborate the merits of the proposed approach.

The rest of this article is organized as follows. Section II introduces the signal model of the long-baseline radar network. In Section III, the joint target detection and localization problem is formulated and solved based on the GLRT and the AMF. Section IV presents the numerical results of experimental simulations. Finally, Section V concludes this article.

Notations: $\|\cdot\|$ represents the 2-norm operator. $E(\cdot)$ denotes mathematical expectation. The symbols $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^\dagger$ represent the transpose, the conjugate, and the conjugate transpose operators, respectively. For the complex number, $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of the numbers in parentheses, respectively. j represents the imaginary unit (i.e., $j = \sqrt{-1}$). $\mathbb{R}^{M \times N}$ denotes real matrices. $\mathbb{C}^{N \times M}$ represents $N \times M$ complex matrices. $\mathbf{I}_N$ and $\mathbf{0}_N$ represent the N -order identity matrix and null matrix, respectively. $\det(\mathbf{A})$ indicates the determinant of the matrix $\mathbf{A}$.

II. SIGNAL MODEL OF THE ACTIVE RADAR NETWORK

The active radar network with N monostatic radars is considered in a space rectangular coordinate system. As shown in Fig. 1, the radars are located at $\mathbf{p}_{R,n}$ and move with velocities $\mathbf{v}_{R,n}$, $n = 1, 2, \dots, N$ ($N > 1$). In the common surveillance region, a moving target is located, whose location and velocity are expressed as $\mathbf{p}$ and $\mathbf{v}$, respectively.

All radars observe the target simultaneously, at least with a very negligible observation time difference. The n th radar transmits M_n pulse signals with the pulse repetition interval (PRI) $T_{r,n}$. The transmitted signal of the n th radar is denoted by

$$y_n(t) = \Re \{ s_n(t) e^{j2\pi f_{c,n}t} \} \quad (1)$$

where $s_n(t)$ represents the baseband signal with the pulse duration τ_n and the bandwidth B_n . $f_{c,n}$ indicates the carrier frequency. All nodes choose different frequency bandwidths

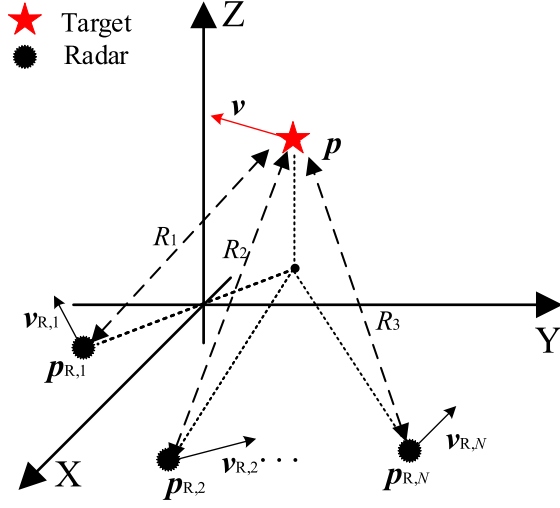


Fig. 1. Spatial configuration.

so that radars will not interfere with each other. Each radar only receives the signal transmitted by itself [17], [33].

Suppose that there is no range and Doppler migration; the signal of the n th node during the m th ($1 \leq m \leq M_n$) pulse is then under the classical narrowband approximation

$$r_{n,m}(t) = \alpha_n s_n(t - t_n) e^{j2\pi f_{d,n}(m-1)T_{r,n}} + w_{n,m}(t) \quad (2)$$

where α_n denotes the complex amplitude, which is assumed to be an unknown deterministic parameter. In this model, the amplitude of the target echo remains invariant during a coherent processing interval. This assumption holds when the amplitude fluctuates slightly. Similar models have been adopted for moving target detection and are discussed widely [24], [55], [56]. The term t_n indicates the time delay trip for the n th radar scattered by the target and is expressed as

$$t_n = t_n(\mathbf{p}) = \frac{2}{c} \sqrt{(\mathbf{p} - \mathbf{p}_{R,n})^T (\mathbf{p} - \mathbf{p}_{R,n})} \quad (3)$$

where c is the speed of light. The term $f_{d,n}$ represents the Doppler frequency shift that is caused by the relative motion between the target and the n th radar

$$\begin{aligned} f_{d,n} &= f_{d,n}(\mathbf{p}, \mathbf{v}) \\ &= \frac{2(\mathbf{p} - \mathbf{p}_{R,n})^T (\mathbf{v} - \mathbf{v}_{R,n})}{\lambda_n \sqrt{(\mathbf{p} - \mathbf{p}_{R,n})^T (\mathbf{p} - \mathbf{p}_{R,n})}} \end{aligned} \quad (4)$$

where $\lambda_n = c/f_{c,n}$ is the wavelength. The term $w_{n,m}(t)$ in (2) is the noise plus clutter, which is modeled as complex Gaussian random processes.

For further process, the signals from different views should be mapped into a common coordinate system, where the common surveillance region of the radar network is subdivided into multiple spatial resolution cells (SRCs), which are the minimum cells for processing [33], so each of SRC can gather observation data from the corresponding fast-time samples. The target information can be extracted by fusion of the observation data corresponding to the SRC.

Given an SRC with the center $\mathbf{p}_c$, the delay difference between the actual target and the SRC is $\Delta t_n(\mathbf{p}) = t_n(\mathbf{p}_c) - t_n(\mathbf{p})$. If an exact knowledge of the target position is available (i.e., $\Delta t_n(\mathbf{p}) = 0$, $n = 1, 2, \dots, N$), then a perfect sampling can be performed, as assumed in [24], [40], and [55]. However, the perfect sampling cannot be implemented, so the mismatched offset usually accompanies the signal corresponding to the SRC

$$r_{n,m}(t) = \alpha_n s_n(\Delta t_n(\mathbf{p})) e^{j2\pi f_{d,n}(m-1)T_{r,n}} + w_{n,m}(t_n(\mathbf{p}_c)). \quad (5)$$

After sampling with T_s , the continuous signal from the SRC can be written in a vector form. For the convenience of analysis, two vectors are defined, i.e., $\mathbf{s}_n(\mathbf{p}) \in \mathbb{C}^{1 \times L_n}$ and $\mathbf{d}_n(\mathbf{p}, \mathbf{v}) \in \mathbb{C}^{M_n \times 1}$

$$\mathbf{s}_n(\mathbf{p}) = \mathbf{s}'_n(\Delta t_n) = [s_{n,1}(\mathbf{p}), s_{n,2}(\mathbf{p}), \dots, s_{n,L_n}(\mathbf{p})]$$

(6)

$$\mathbf{d}_n(\mathbf{p}, \mathbf{v}) = \mathbf{d}'_n(f_{d,n}) = \begin{bmatrix} d_{n,1}(\mathbf{p}, \mathbf{v}) \\ d_{n,2}(\mathbf{p}, \mathbf{v}) \\ \vdots \\ d_{n,M_n}(\mathbf{p}, \mathbf{v}) \end{bmatrix} \quad (7)$$

where $s_{n,l}(\mathbf{p}) = s'_{n,l}(\Delta t_n) = s_n(lT_s - \Delta t_n(\mathbf{p}))$ is the l th entry of $\mathbf{s}_n$. $d_{n,m}(\mathbf{p}, \mathbf{v}) = d'_{n,m}(f_{d,n}) = e^{j2\pi f_{d,n}(\mathbf{p}, \mathbf{v})(m-1)T_{r,n}}$ is the m th entry of $\mathbf{d}_n(f_{d,n})$. L_n denotes the length of the n th radar's signal.

Then, the discrete form of the received signal of node n is written as

$$\mathbf{Z}_n = \alpha_n \mathbf{s}_n(\mathbf{p}) \otimes \mathbf{d}_n(\mathbf{p}, \mathbf{v}) + \mathbf{W}_n \quad (8)$$

where $\mathbf{Z}_n = [\mathbf{z}_{n,1}, \mathbf{z}_{n,2}, \dots, \mathbf{z}_{n,L_n}] \in \mathbb{C}^{M_n \times L_n}$ and $\mathbf{W}_n = [\mathbf{w}_{n,1}, \mathbf{w}_{n,2}, \dots, \mathbf{w}_{n,L_n}] \in \mathbb{C}^{M_n \times L_n}$. The term $\mathbf{W}_n$ denotes the noise plus clutter. For narrowband assumption, the clutter samples from different range resolution cells are independent. Thus, the entries are zero-mean complex Gaussian valuables with

$$\mathbb{E}\{\mathbf{w}_{n,l} \mathbf{w}_{n',l'}^H\} = \begin{cases} \mathbf{C}_n, & n = n' \\ \mathbf{0}_{L_n}, & \text{others} \end{cases} \quad (9)$$

$$\mathbf{C}_n = \begin{bmatrix} \sigma_{n,1,1} & \sigma_{n,1,2} & \cdots & \sigma_{n,1,M_n} \\ \sigma_{n,2,1} & \sigma_{n,2,2} & \cdots & \sigma_{n,2,M_n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n,M_n,1} & \sigma_{n,M_n,2} & \cdots & \sigma_{n,M_n,M_n} \end{bmatrix}. \quad (10)$$

The covariance matrix $\mathbf{C}_n \in \mathbb{C}^{M_n \times M_n}$ is the positive-definite covariance matrix of the noise plus clutter in a range cell. All fast-time samples are mutually independent.

This article investigates joint adaptive target detection, position determination, and velocity estimation within the context of Gaussian interference. Therefore, α_n , $\mathbf{p}$, $\mathbf{v}$, and $\sigma_{n,m,m'}$ are considered as the unknowns. For convenience of analysis, we assume that only one target of interest exists, because even if multitarget detection and localization is involved, the potential targets can be extracted one by one using the proposed architecture with the CLEAN algorithm [50].

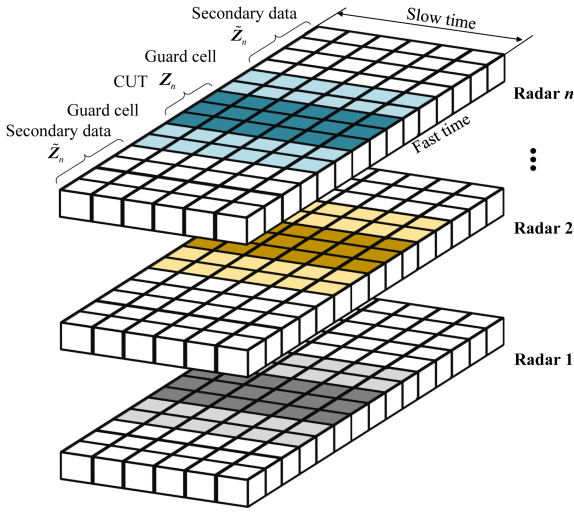


Fig. 2. Probe cell partitioning.

III. JOINT ADAPTIVE TARGET DETECTION AND LOCALIZATION

This section investigates the problem of joint target detection and localization within a given SRC. Both subproblems, i.e., detection and parameter estimation, are addressed in a jointly optimum manner.

Fig. 2 illustrates parts of signal matrices corresponding to the SRC, where $\mathbf{Z}_n \in \mathbb{C}^{M_n \times L_n}$ represents the data of cell under test (CUT). They are the fast-time samples gathered by the SRC from all radar stations. The secondary data $\tilde{\mathbf{Z}}_n = [\tilde{\mathbf{z}}_{n,1}, \tilde{\mathbf{z}}_{n,2}, \dots, \tilde{\mathbf{z}}_{n,K_n}] \in \mathbb{C}^{M_n \times K_n}$, $k = 1, 2, \dots, K_n \geq M_n$, are available to estimate spectral characteristics of the noise plus clutter in the n th radar, which is around the CUT. Usually, some guard cells are added between the CUT and $\tilde{\mathbf{Z}}_n$ to prevent $\tilde{\mathbf{Z}}_n$ from containing the data from the CUT, which will affect the estimation of the spectral characteristics of the noise plus clutter. Hence, the detection problem is formed as a binary hypothesis testing, where H_0 denotes the case of target absence and H_1 denotes target presence within the SRC

$$\begin{aligned} H_0 : & \begin{cases} \mathbf{Z}_n = \mathbf{W}_n \\ \tilde{\mathbf{Z}}_n = \tilde{\mathbf{W}}_n \end{cases} \\ H_1 : & \begin{cases} \mathbf{Z}_n = \alpha_n \mathbf{s}_n(\mathbf{p}) \otimes \mathbf{d}_n(\mathbf{p}, \mathbf{v}) + \mathbf{W}_n \\ \tilde{\mathbf{Z}}_n = \tilde{\mathbf{W}}_n \end{cases} \\ n = & 1, 2, \dots, N \end{aligned} \quad (11)$$

where $\tilde{\mathbf{W}}_n = [\tilde{\mathbf{w}}_{n,1}, \tilde{\mathbf{w}}_{n,2}, \dots, \tilde{\mathbf{w}}_{n,K_n}] \in \mathbb{C}^{M_n \times K_n}$. The vectors $\tilde{\mathbf{w}}_{n,k}$ and $\mathbf{w}_{n,l}$ are identically modeled as independent distributed random vectors, i.e., $\tilde{\mathbf{w}}_{n,k}, \mathbf{w}_{n,l} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}_n)$.

The joint probability density functions (PDFs) of the n th radar's observations under H_0 can be written as

$$\begin{aligned} f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \mathbf{C}_n, H_0) \\ = \left(\frac{1}{\pi^{M_n} \det(\mathbf{C}_n)} e^{-\text{tr}(\mathbf{C}_n^{-1} \mathbf{T}_n^0)} \right)^{K_n + L_n} \end{aligned} \quad (12)$$

where the term $\mathbf{T}_n^0$ is defined as

$$\mathbf{T}_n^0 = \frac{\sum_{l=1}^{L_n} \mathbf{z}_{n,l} \mathbf{z}_{n,l}^\dagger + \sum_{k=1}^{K_n} \tilde{\mathbf{z}}_{n,k} \tilde{\mathbf{z}}_{n,k}^\dagger}{K_n + L_n}. \quad (13)$$

Similarly, the PDF under H_1 can be expressed as

$$\begin{aligned} f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \mathbf{C}_n, \alpha_n, \mathbf{p}, \mathbf{v}, H_1) \\ = \left(\frac{1}{\pi^{M_n} \det(\mathbf{C}_n)} e^{-\text{tr}(\mathbf{C}_n^{-1} \mathbf{T}_n^1)} \right)^{K_n + L_n} \end{aligned} \quad (14)$$

where $\mathbf{T}_n^1$ is defined as

$$\mathbf{T}_n^1 = \frac{\sum_{l=1}^{L_n} \mathbf{z}'_{n,l} \mathbf{z}'_{n,l}^\dagger + \sum_{k=1}^{K_n} \tilde{\mathbf{z}}_{n,k} \tilde{\mathbf{z}}_{n,k}^\dagger}{K_n + L_n} \quad (15)$$

$$\mathbf{z}'_{n,l} = \mathbf{z}_{n,l} - \alpha_n \mathbf{s}_{n,l}(\mathbf{p}) \mathbf{d}_n(\mathbf{p}, \mathbf{v}). \quad (16)$$

Although the LRT can be seen as the optimal detector and estimator for the problem [11], it cannot be implemented because the perfect knowledge of α_n and $\mathbf{C}_n$ ($n = 1, 2, \dots, N$) is required. The estimation of the interference and target parameters (i.e., $\alpha_1, \alpha_2, \dots, \alpha_N, \mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_N, \mathbf{p}, \mathbf{v}$) is a prerequisite for an implementable radar network detector or estimator. Thus, GLRT- and AMF-based architectures are proposed to design the detector/estimator.

A. GLRT-Based Method

In this section, the GLRT criterion is applied to the detection and estimation problem. Given that the signals received from different radars are independent of each other, the decision statistic can be expressed as

$$\frac{\max_{\alpha_1, \dots, \alpha_N, \mathbf{C}_1, \dots, \mathbf{C}_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \mathbf{C}_n, \alpha_n, \mathbf{p}, \mathbf{v}, H_1)}{\max_{\mathbf{C}_1, \dots, \mathbf{C}_N} \prod_{n=1}^N f_n(\mathbf{Z}_n, \tilde{\mathbf{Z}}_n | \mathbf{C}_n, H_0)} \underset{H_0}{\overset{H_1}{\geq}} \xi \quad (17)$$

where ξ denotes the detection threshold determined by the false alarm probability.

According to [51], (17) can be proceeded with GLRT [the AMF is seen in (37)]. Both the numerator and the denominator are maximized with respect to the covariance matrices, $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_N$. Therefore, the GLRT can be written as

$$\frac{\prod_{n=1}^N \det(\mathbf{T}_n^0)^{K_n + L_n}}{\min_{\alpha_1, \alpha_2, \dots, \alpha_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N \det(\mathbf{T}_n^1)^{K_n + L_n}} \underset{H_0}{\overset{H_1}{\geq}} \xi. \quad (18)$$

Then, denoted by $\mathbf{S}_n = \sum_{k=1}^{K_n} \tilde{\mathbf{z}}_{n,k} \tilde{\mathbf{z}}_{n,k}^\dagger \in \mathbb{C}^{M_n \times M_n}$, (18) can be further simplified as

$$\frac{\prod_{n=1}^N \left(\sum_{l=1}^{L_n} \mathbf{z}_{n,l} \mathbf{z}_{n,l}^\dagger \mathbf{S}_n^{-1} \mathbf{z}_{n,l} + 1 \right)^{K_n + L_n}}{\min_{\alpha_1, \alpha_2, \dots, \alpha_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N \left(\sum_{l=1}^{L_n} \mathbf{z}'_{n,l} \mathbf{z}'_{n,l}^\dagger \mathbf{S}_n^{-1} \mathbf{z}'_{n,l} + 1 \right)^{K_n + L_n}} \underset{H_0}{\overset{H_1}{\geq}} \xi \quad (19)$$

where $\mathbf{S}_n$ is proportional to the sampling covariance matrix of the n th radar via K_n .

Now, maximizing this quantity over $\alpha_1, \alpha_2, \dots, \alpha_N$. With several derivations (also seen in Appendix A), the

decision rule boils down to

$$\max_{\mathbf{p}, \mathbf{v}} \prod_{n=1}^N \left(1 - \frac{q_n(\mathbf{p}, \mathbf{v})}{p_n} \right)^{-(K_n + L_n)} \underset{H_0}{\overset{H_1}{\geq}} \xi \quad (20)$$

where the term $p_n = 1 + \sum_{l=1}^{L_n} \mathbf{z}_{n,l}^\dagger \mathbf{S}_n^{-1} \mathbf{z}_{n,l}$ is not related to $\mathbf{p}$ and $\mathbf{v}$. The term $q_n(\mathbf{p}, \mathbf{v})$ that refers to the unknown parameters is defined as

$$\begin{aligned} q_n(\mathbf{p}, \mathbf{v}) &= q'_n(\Delta t_n, f_{d,n}) \\ &= \frac{\left| \sum_{l=1}^{L_n} s_{n,l}^* (\mathbf{p}) \mathbf{d}_n^\dagger (\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{z}_{n,l} \right|^2}{\sum_{l=1}^{L_n} |s_{n,l} (\mathbf{p})|^2 \mathbf{d}_n^\dagger (\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{d}_n (\mathbf{p}, \mathbf{v})}. \end{aligned} \quad (21)$$

The relationship between the detection threshold and the probability of false alarm (P_{fa}) can be established by Monte Carlo (MC) experiments. This work can be done offline, ensuring no impact on the real-time performance of the radar network. Likewise, we can derive the detection performance curve for the global detector via MC experiments, as illustrated in Fig. 4. As the deviation between the detector's input and the target diminishes, the detection performance progressively improves. However, when the deviation becomes sufficiently small, the enhancement in detection performance becomes negligible, gradually approaching the theoretical performance upper bound. Consequently, the detection results with the known target's position and velocity can serve as the optimal detection performance boundary, which remains unattainable in practical applications.

After a logarithm transform, the GLRT statistic can be expressed as

$$\Lambda_{\text{GLRT}} = \max_{\mathbf{p}, \mathbf{v}} \phi_{\text{GLRT}}(\mathbf{p}, \mathbf{v}) \quad (22)$$

where the function $\phi_{\text{GLRT}}(\mathbf{p}, \mathbf{v})$ is defined as

$$\phi_{\text{GLRT}}(\mathbf{p}, \mathbf{v}) = - \sum_{n=1}^N (K_n + L_n) \ln \left(1 - \frac{q_n(\mathbf{p}, \mathbf{v})}{p_n} \right). \quad (23)$$

When the decision statistic exceeds the detection threshold ξ , the target's position and velocity can be obtained from the decision statistic computation (20). Thus, we can handle detection and estimation problems as follows:

$$\begin{aligned} &\max_{\mathbf{p}, \mathbf{v}} \phi_{\text{GLRT}}(\mathbf{p}, \mathbf{v}) \\ &\text{s.t. } \Lambda_{\text{GLRT}} > \xi. \end{aligned} \quad (24)$$

Note that (24) involves a nonconvex nonlinear optimization problem. The iterative or grid search method is practically used to solve the problem. For grid search methods, target detection and localization performance are subject to the grid size. We must search for the optimal solution with dense grids to obtain a satisfactory performance. That may cause great computing complexity and a possible straddle loss. The iterative method needs a good initial starting point, which can be provided by indefinite grid search or the prediction of a tracker. We assume that the rough solution with inaccurate offset is provided for convenience. Based on

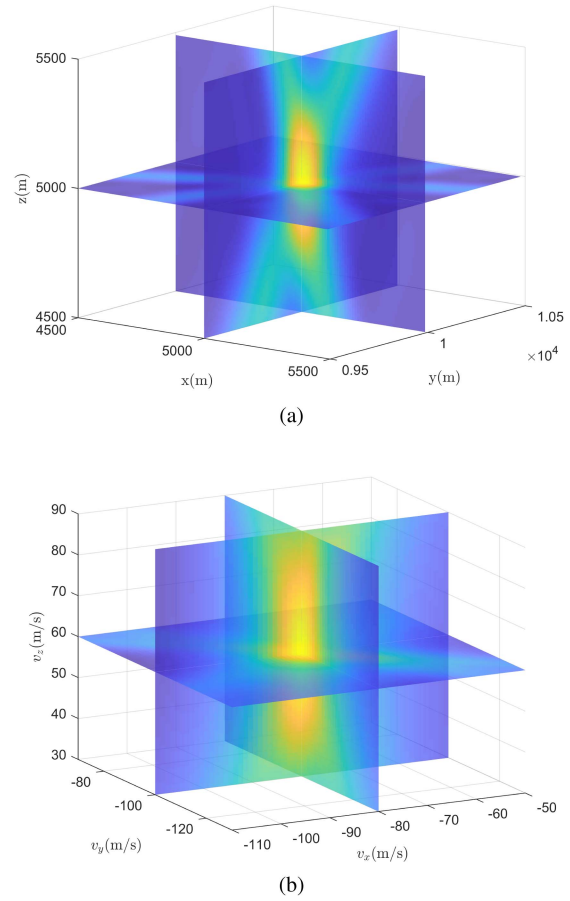


Fig. 3. (a) and (b) Objective function when the SCNR = 15 dB.

the iterative method, two refinement methods are proposed, which are considered as follows:

- 1) a direct refinement strategy based on the steepest ascent method initialized by the given rough point;
- 2) a mixed refinement strategy where the indirect estimation obtains an indefinite solution, and then, the aforementioned direct refinement is initialized with an output of the indirect method.

1) *Direct Refinement Strategy*: This section considers a direct refinement strategy based on the gradient method. As shown in Fig. 3, the objective function (23) can be seen as the spatial power function. The spatial power is concentrated on the SRC containing a target. The local maximum value within the SRC is the position of the target when the signal-to-clutter-plus-noise ratio (SCNR) is high enough. In order to obtain the local maximum value based on the gradient method, the starting initial point must be within the SRC

$$|\Delta t_n(\mathbf{p}) - \Delta t_n(\mathbf{p}_0)| \leq \frac{1}{2B_n} \quad (25)$$

and

$$|f_{d,n}(\mathbf{p}, \mathbf{v}) - f_{d,n}(\mathbf{p}_0, \mathbf{v}_0)| \leq \frac{1}{2M_n T_{r,n}}. \quad (26)$$

We can use the center of SRC as the initial starting point, which can be obtained by the grid search or the prediction

of the tracker. For the sake of analysis, the initial starting point is assumed to be given.

When the given point meets the conditions (25) and (26), the problem (24) can be locally convex. In this situation, some iterative procedures can be applied to the optimization problem, such as the Newton and steepest ascent methods. The Newton method works with a high computational complexity because of the calculation of the Hessian matrix. The low complexity of the steepest ascent-based method can provide a satisfactory performance, which is employed for the refinement procedure. The recursion formula is given as

$$\begin{aligned} \mathbf{p}_g &= \mathbf{p}_{g-1} + \mu \Delta \mathbf{p} \\ \mathbf{v}_g &= \mathbf{v}_{g-1} + \mu \Delta \mathbf{v} \end{aligned} \quad (27)$$

where $\mathbf{p}_g$ and $\mathbf{v}_g$ are outputs of the g th iteration. $\mu > 0$ represents the step factor.

From (27), the recursion formula contains the gradient and the step factor. Depending on the chain rule, the gradient of the composite function is obtained by

$$\begin{aligned} \Delta \mathbf{p} &= \frac{\partial \phi_{\text{GLRT}}(\mathbf{p}, \mathbf{v})}{\partial \mathbf{p}} \\ &= \sum_{n=1}^N \frac{(K_n + 1)}{p_n - q_n(\mathbf{p}, \mathbf{v})} \frac{\partial q'_n(\Delta t_n, f_{d,n})}{\partial f_{d,n}} \frac{\partial f_{d,n}}{\partial \mathbf{p}} \\ &\quad + \sum_{n=1}^N \frac{(K_n + L_n)}{p_n - q_n(\mathbf{p}, \mathbf{v})} \frac{\partial \tilde{q}_n(\Delta t_n, f_{d,n})}{\partial \Delta t_n} \frac{\partial \Delta t_n}{\partial \mathbf{p}} \end{aligned} \quad (28)$$

$$\begin{aligned} \Delta \mathbf{v} &= \frac{\partial \phi_{\text{GLRT}}(\mathbf{p}, \mathbf{v})}{\partial \mathbf{v}} \\ &= \sum_{n=1}^N \frac{(K_n + L_n)}{p_n - q_n(\mathbf{p}, \mathbf{v})} \frac{\partial q'_n(\Delta t_n, f_{d,n})}{\partial f_{d,n}} \frac{\partial f_{d,n}}{\partial \mathbf{v}} \end{aligned} \quad (29)$$

where the terms $\partial q_n(\mathbf{p}, \mathbf{v})/\partial \mathbf{p}$ and $\partial q_n(\mathbf{p}, \mathbf{v})/\partial \mathbf{v}$ are the partial derivatives. Interested readers can refer to the detailed expressions in Appendix B.

The step factor is obtained by an inaccurate 1-D search method. We consider the search method based on the golden section. Because the objective function is not globally convex, the initial search range should be constrained to meet (25) and (26)

$$|\Delta t_n(\mathbf{p} + \mu \Delta \mathbf{p}) - \Delta t_n(\mathbf{p}_0)| \leq \frac{1}{2B_n} \quad (30)$$

$$|f_{d,n}(\mathbf{p} + \mu \Delta \mathbf{p}, \mathbf{v} + \mu \Delta \mathbf{v}) - f_{d,n}(\mathbf{p}_0, \mathbf{v}_0)| \leq \frac{1}{2M_n T_{r,n}}. \quad (31)$$

By expanding the left-hand side of (30) and (31) in the Taylor series and only keeping a first-order expression, the inequality set can be simplified into

$$|\mu| \leq \min \left\{ \left| \frac{1}{2B_n \frac{\partial \Delta t_n}{\partial \mathbf{p}} \Delta \mathbf{p}} \right|, \left| \frac{1}{2M_n T_{r,n}} \left(\Delta \mathbf{p} \frac{\partial f_{d,n}}{\partial \mathbf{p}} + \Delta \mathbf{v} \frac{\partial f_{d,n}}{\partial \mathbf{v}} \right) \right| \right\}. \quad (32)$$

Algorithm 1: Search Method for the Step Factor.

Input: $\Delta \mathbf{p}, \Delta \mathbf{v}, \varepsilon'$

Output: μ

```

1: obtain  $\mu_{\max}$  by substituting  $\Delta \mathbf{p}$  and  $\Delta \mathbf{v}$  into (32)
2:  $a = 0, b = \mu_{\max}$ ;
3: while  $b - a > \varepsilon'$  do
4:    $x_1 = a + 0.382(b - a), x_2 = a + 0.618(b - a)$ ;
5:   if  $\phi_{\text{GLRT}}(\mathbf{p} + x_1 \Delta \mathbf{p}, \mathbf{v} + x_2 \Delta \mathbf{v}) >$ 
      $\phi_{\text{GLRT}}(\mathbf{p} + x_2 \Delta \mathbf{p}, \mathbf{v} + x_1 \Delta \mathbf{v})$  then
6:      $b = x_2$ ;
7:   else
8:      $a = x_1$ ;
9:   end if
10: end while
11:  $\mu = (a + b)/2$ 
12: return  $\mu$ ;
```

Algorithm 2: Detection and Localization With Direct Refinement Procedure.

Input: $G, \varepsilon_1, \varepsilon_2, \mathbf{p}_0, \mathbf{v}_0, P_{\text{fa}}$

Output: $\hat{\mathbf{p}}, \hat{\mathbf{v}}$

```

1: for  $g = 0 : G$  do
2:   obtain  $\Delta \mathbf{p}$  and  $\Delta \mathbf{v}$  by (28) and (29);
3:   obtain  $\mu$  using Algorithm 1;
4:    $\mathbf{p}_g = \mathbf{p}_{g-1} + \mu \Delta \mathbf{p}, \mathbf{v}_g = \mathbf{v}_{g-1} + \mu \Delta \mathbf{v}$ ;
5:   if  $|\Delta \mathbf{p}| + |\Delta \mathbf{v}| < \varepsilon_1$  then
6:     if  $\phi_{\text{GLRT}}(\mathbf{p}_g, \mathbf{v}_g) < \xi(P_{\text{fa}})$  then
7:        $\mathbf{p}_g = \text{null}, \mathbf{v}_g = \text{null}$ ;
8:       break;
9:     end if
10:   end if
11:   if  $|\Delta \mathbf{p}| + |\Delta \mathbf{v}| < \varepsilon_2$  then
12:     break;
13:   end if
14: end for
15:  $\hat{\mathbf{p}} = \mathbf{p}_g, \hat{\mathbf{v}} = \mathbf{v}_g$ ;
16: return  $\hat{\mathbf{p}}, \hat{\mathbf{v}}$ ;
```

Therefore, the search method for the step factor can be seen in Algorithm 1.

As for the stopping condition, the regular method is determining $\hat{\mathbf{p}}$ and $\hat{\mathbf{v}}$ applying (27) until a maximum iteration number G or the required accuracy ε is achieved. If the statistic $\phi_{\text{GLRT}}(\mathbf{p}_g, \mathbf{v}_g)$ exceeds the threshold, $\mathbf{p}_g$ and $\mathbf{v}_g$ are output as the final estimates of the physical target. However, this method has unnecessary iterations under the absence (H_0) hypothesis.

In fact, target detection does not require high accuracy and a large iteration number. As shown in Fig. 4, when the required accuracy ε is small enough, the detection performance only obtains a negligible improvement with ε increasing. Thus, we take two required accuracies for our proposed framework. The first required accuracy ε_1 is set for target detection. When this accuracy is achieved, the statistic is compared with the detection threshold $\xi(P_{\text{fa}})$. If the statistic exceeds the threshold, the further iteration will

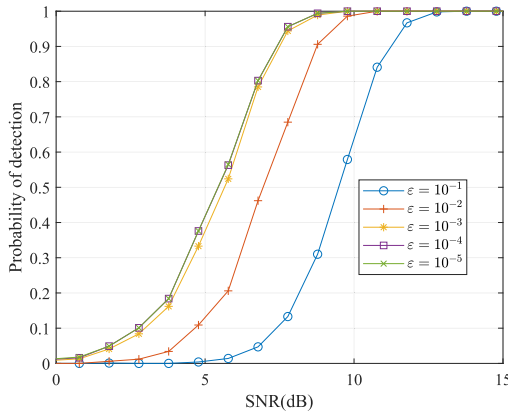


Fig. 4. Detection probability at different ε .

be implemented. The estimates are output until the second required accuracy ε_2 is achieved. When the target is absent, unnecessary iterations are saved.

Finally, the detection and localization with direct refinement procedure is provided in Algorithm 2. Unlike the regular method, the proposed method adopts a two-step strategy, where the statistic with a rough solution is first obtained for detection. Then, further iteration is implemented for exact estimates if the statistic exceeds the detection threshold. In this way, further refinement will be canceled under the H_0 hypothesis. Algorithm 2 is proposed to trade off computational complexity and detection performance.

Based on the aforementioned refinement strategy, the expression of the GLRT-based detector can be expressed as

$$\phi_{\text{GLRT}}(\hat{\mathbf{p}}, \hat{\mathbf{v}}) \underset{H_0}{\overset{H_1}{\geq}} \xi \quad (33)$$

where $\hat{\mathbf{p}}$ and $\hat{\mathbf{v}}$ indicate the estimates of position and velocity, respectively, which are provided by the refinement algorithms.

The computational complexity of grid partitioning lies in the calculation of sample covariance matrix $\mathbf{C}_1, \dots, \mathbf{C}_N$ and the number of grids G in the common surveillance region. The computational complexity of the sample covariance matrix is $\mathcal{O}(\sum_{n=1}^N K_n M_n^2)$, so the computational complexity of grid partitioning is $\mathcal{O}(G \sum_{n=1}^N K_n M_n^2)$. The computational complexity for the direct refinement procedure based on Algorithm 2 lies in the calculation of sample covariance matrix $\mathbf{C}_1, \dots, \mathbf{C}_N$ and the iteration number N_{it} , $\mathcal{O}(\sum_{n=1}^N K_n M_n^2 + N_{\text{it}})$.

In order to guarantee the convergence of the algorithms, the offset with respect to the nominal position and velocity should be small enough to meet (25) and (26). When the technique is applied to a search task, the initial starting point is provided by the grid search method. The direct refinement method can search for the optimal solution with sparse grids to obtain satisfactory performance. However, it is essential to limit the grid's size to ensure that the initial starting point obtained through the grid search adheres to the necessary constraints. In tracking tasks, predicted parameters provided by the tracker are used as the initial starting

point. This approach relies heavily on tracking accuracy. The tracker must deliver high precision to ensure that the starting point meets the required constraints. If the offset of the initial starting point exceeds acceptable limits, the optimization problem in (24) may lose its local convexity. This scenario can lead to significant errors in estimating position and velocity, undermining the overall effectiveness of the optimization process.

2) Mixed Refinement Strategy: In this section, we consider a mixed refinement strategy, where Algorithm 2 is initialized by outputs of an indirect method. The direct refinement strategy can only provide good performance in the strict condition of (25) and (26). On the contrary, the indirect method allows a larger error between the grid point and the physical target but cannot achieve the optimum solution, where the position and velocity estimates are calculated by solving several equations. In order to make full use of the advantages of the direct method and the indirect method, the mixed refinement strategy is proposed, where the direct refinement is initialized by the outputs of the indirect method. In this way, a satisfactory performance of detection and localization can be obtained, while a larger error is allowed.

First, the time delay and Doppler frequency are extracted on each signal separately, and then, the precise position and velocity are given by solving several equations. The time delay and the Doppler frequency are obtained only using the local statistic

$$[\hat{\Delta t}_n, \hat{f}_{d,n}] = \arg \max_{\Delta t_n, f_{d,n}} q'_n(\Delta t_n, f_{d,n}) \quad (34)$$

where $\hat{\Delta t}_n$ and $\hat{f}_{d,n}$ are the maximum likelihood estimates (MLEs). A reasonable way to approximate the maximization in (34) is to search in a finite discretized delay-Doppler space. The performance of the indirect method is limited by the quantization error. Further refinement proceeding is implemented after the rough indirect estimation.

Then, the position and velocity estimates are given by solving the following overdetermined equation set:

$$\begin{cases} \Delta t_n(\mathbf{p}) = \hat{\Delta t}_n \\ f_{d,n}(\mathbf{p}, \mathbf{v}) = \hat{f}_{d,n} \end{cases}, \quad n = 1, 2, \dots, N. \quad (35)$$

We solve (35) based on a weighted least squares criterion, where the cost function is given using the local estimates [43]

$$\begin{aligned} [\hat{\mathbf{p}}_0, \hat{\mathbf{v}}_0] = \arg \max_{\mathbf{p}, \mathbf{v}} & \sum_{n=1}^N w_{t,n} |\Delta t_n(\mathbf{p}) - \hat{\Delta t}_n|^2 \\ & + \sum_{n=1}^N w_{f,n} |f_{d,n}(\mathbf{p}, \mathbf{v}) - \hat{f}_{d,n}|^2 \end{aligned} \quad (36)$$

where $w_{t,n}$ and $w_{f,n}$ indicate the weight, whose value is the inverse of CRLB on range-Doppler [see (73) and (75)]. By solving (36) iteratively, we can obtain the indirect estimates of $\hat{\mathbf{p}}_0$ and $\hat{\mathbf{v}}_0$.

Finally, by using the results, $\hat{\mathbf{p}}_0$ and $\hat{\mathbf{v}}_0$, as the starting initialized point, the aforementioned direct refinement is

implemented to obtain the final detection and localization results.

The mixed refinement method is a modified version of the direct one, demonstrating greater robustness in scenarios with substantial offsets between the initialized point and the peak. When applied to search tasks, the mixed refinement method can work with sparser grids compared to the direct method, leading to reduced complexity. In tracking tasks, the mixed method can work with large tracking errors, providing more robustness. However, under low-signal-to-noise ratio (SNR) conditions, the errors of the indefinite solution obtained by the indirect estimation become significant, which will degrade the performance.

B. AMF-Based Method

This section applies the AMF criterion to design the detector (or estimator). First, the covariance matrices, $\mathbf{C}_1, \dots, \mathbf{C}_N$, are assumed to be known. Then, GLRT is derived over the remaining unknown $\alpha_1, \dots, \alpha_N, \mathbf{p}$, and $\mathbf{v}$. Finally, the covariance matrices are replaced by the sample covariance matrix, which is estimated using the secondary data $\tilde{\mathbf{Z}}_1, \dots, \tilde{\mathbf{Z}}_N$. The decision rule based on the AMF is given as

$$\frac{\max_{\alpha_1, \dots, \alpha_N, \mathbf{p}, \mathbf{v}} \prod_{n=1}^N f_n(\mathbf{Z}_n | \mathbf{C}_n = \mathbf{K}_n^{-1} \mathbf{S}_n, \alpha_n, \mathbf{p}, \mathbf{v}, H_1)}{\prod_{n=1}^N f_n(\mathbf{Z}_n | \mathbf{C}_n = \mathbf{K}_n^{-1} \mathbf{S}_n, H_0)} \underset{H_0}{\overset{H_1}{\geq}} \zeta \quad (37)$$

where the threshold ζ can also be denoted by $\zeta(P_{fa})$. The likelihood function in (37) are, respectively, given by

$$\begin{aligned} f_n(\mathbf{Z}_n | \mathbf{C}_n = \mathbf{K}_n^{-1} \mathbf{S}_n, \alpha_n, \mathbf{p}, \mathbf{v}, H_1) \\ = D e^{-\sum_{l=1}^{L_n} \mathbf{z}_{n,l}'^* \mathbf{S}_n^{-1} \mathbf{z}_{n,l}'} \end{aligned} \quad (38)$$

and

$$f_n(\mathbf{Z}_n | \mathbf{C}_n = \mathbf{K}_n^{-1} \mathbf{S}_n, H_0) = D e^{-\sum_{l=1}^{L_n} \mathbf{z}_{n,l}'^* \mathbf{S}_n^{-1} \mathbf{z}_{n,l}'} \quad (39)$$

where D is a constant related to $\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_N$. The term $\mathbf{z}_{n,l}'$ is defined in (16). Note that $\alpha_1, \alpha_2, \dots, \alpha_N$ are unknown parameters. For estimating α_n , with a logarithm operation, the likelihood function can be simplified into

$$\begin{aligned} \ln f_n(\mathbf{Z}_n | \alpha_n, \mathbf{p}, \mathbf{v}, H_1) &\propto - \sum_{l=1}^{L_n} \mathbf{z}_{n,l}'^* \mathbf{S}_n^{-1} \mathbf{z}_{n,l}' \\ &= 2 \sum_{l=1}^{L_n} \Re \{ \alpha_n^* s_{n,l}^* \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{z}_{n,l} \} \\ &\quad - \sum_{l=1}^{L_n} |\alpha_n s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{z}_{n,l} \\ &\quad - \sum_{l=1}^{L_n} \mathbf{z}_{n,l}'^* \mathbf{S}_n^{-1} \mathbf{z}_{n,l}'. \end{aligned} \quad (40)$$

Maximizing over $\alpha_1, \dots, \alpha_N$ under H_1 hypothesis yields

$$\hat{\alpha}_n = \frac{\sum_{l=1}^{L_n} s_{n,l}^* \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{z}_{n,l}}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \quad (41)$$

where $\hat{\alpha}_n$ represents the MLE of complex amplitude with $n = 1, 2, \dots, N$. The derivation can be seen in Appendix A.

Substituting $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_N$ into (37), the decision rule is expressed as

$$\max_{\mathbf{p}, \mathbf{v}} \sum_{n=1}^N \frac{\left| \sum_{l=1}^{L_n} s_{n,l}^* \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{z}_{n,l} \right|^2}{\sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})} \underset{H_0}{\overset{H_1}{\geq}} \zeta. \quad (42)$$

Therefore, the statistic for both detection and estimation is given as

$$\Lambda_{\text{AMF}} = \max_{\mathbf{p}, \mathbf{v}} \sum_{n=1}^N q_n(\mathbf{p}, \mathbf{v}) \quad (43)$$

where the function $q_n(\mathbf{p}, \mathbf{v})$ is defined in (21).

Similarly, the detection and estimation problem can be expressed as

$$\begin{aligned} \max_{\mathbf{p}, \mathbf{v}} \phi_{\text{AMF}}(\mathbf{p}, \mathbf{v}) \\ \text{s.t. } \Lambda_{\text{AMF}} > \zeta \end{aligned} \quad (44)$$

where ϕ_{AMF} is obtained by

$$\phi_{\text{AMF}} = \sum_{n=1}^N q_n(\mathbf{p}, \mathbf{v}). \quad (45)$$

Unlike GLRT, the gradient of AMF can be obtained as

$$\begin{aligned} \Delta \mathbf{p} &= \frac{\partial \phi_{\text{AMF}}(\mathbf{p}, \mathbf{v})}{\partial \mathbf{p}} \\ &= \sum_{n=1}^N \frac{\partial q_n'(\Delta t_n, f_{d,n})}{\partial f_{d,n}} \frac{\partial f_{d,n}}{\partial \mathbf{p}} \\ &\quad + \sum_{n=1}^N \frac{\partial q_n'(\Delta t_n, f_{d,n})}{\partial \Delta t_n} \frac{\partial \Delta t_n}{\partial \mathbf{p}} \end{aligned} \quad (46)$$

$$\begin{aligned} \Delta \mathbf{v} &= \frac{\partial \phi_{\text{AMF}}(\mathbf{p}, \mathbf{v})}{\partial \mathbf{v}} \\ &= \sum_{n=1}^N \frac{\partial q_n'(\Delta t_n, f_{d,n})}{\partial f_{d,n}} \frac{\partial f_{d,n}}{\partial \mathbf{v}} \end{aligned} \quad (47)$$

where the partial derivatives $\partial q_n(\mathbf{p}, \mathbf{v})/\partial \mathbf{p}$ and $\partial q_n(\mathbf{p}, \mathbf{v})/\partial \mathbf{v}$ are derived in Appendix B. Equation (43) can be solved in the same way as (24). Algorithm 2 is also applied to the AMF design. As a consequence, the AMF detector with the refinement procedure is expressed as

$$\phi_{\text{AMF}}(\hat{\mathbf{p}}, \hat{\mathbf{v}}) \underset{H_0}{\overset{H_1}{\geq}} \zeta. \quad (48)$$

IV. NUMERICAL EXPERIMENT

In this section, we compare the proposed strategies with some corresponding strategies in the public literature to evaluate their performance. In order to prove the validity of the proposed adaptive architectures, 2-D and 3-D scenarios are considered.

The covariance matrix of echo is modeled as $\mathbf{C}_n = \mathbf{C}_{c,n} + \sigma_n^2 \mathbf{I}$, where σ_n^2 is the power of white noise and $\mathbf{C}_{c,n}$

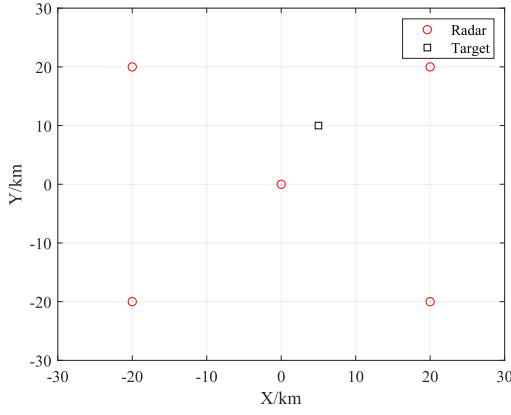


Fig. 5. Placement of radar nodes in the 2-D scenario.

refers to the clutter covariance. The detection performance is evaluated by the probability of target detection over a CFAR. Moreover, the threshold corresponds to the probability of false alarm $P_{fa} = 10^{-3}$ by $100/P_{fa}$ MC realizations. Regarding the estimation performance, the RMSE is considered the figure of merit. The probability of detection and the RMSE are provided, resorting to 1000 MC runs. Finally, given that transmitted signals are satisfied $\mathbf{s}_n^\dagger \mathbf{s}_n = 1$, the SCNR is defined as

$$\text{SCNR} = \frac{1}{N} \sum_{n=1}^N |\alpha_n|^2 \mathbf{d}_n^\dagger (\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{d}_n (\mathbf{p}, \mathbf{v}). \quad (49)$$

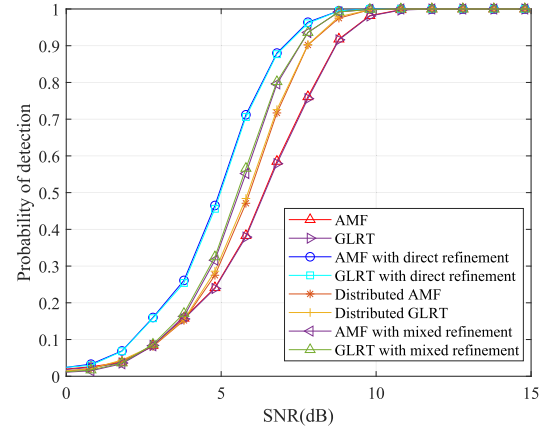
For analyzing the detector's performance, the decision statistics referred to as GLRT [40], AMF [24], and their distributed forms, are compared. GLRT and AMF detectors treat the nominal position $\mathbf{p}_C$ as useful target parameters (they commonly operate in mismatched conditions and are thus also referred to as mismatched detectors). As for the distributed detectors, in the following, the distributed GLRT detector is

$$\Lambda_{\text{DGLRT}} = - \sum_{n=1}^N (K_n + L_n) \ln \left(1 - \frac{\max_{\Delta t_n, f_n} \tilde{q}_n(\Delta t_n, f_{d,n})}{p_n} \right). \quad (50)$$

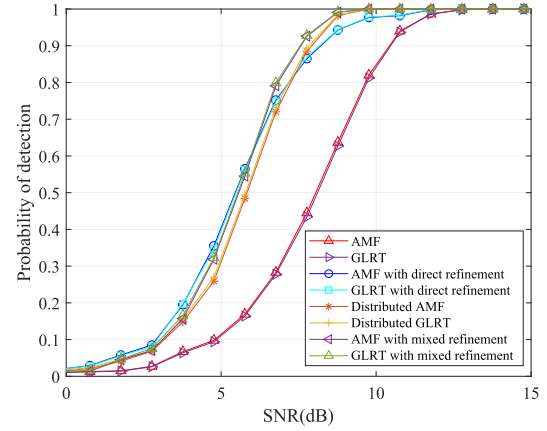
Similarly, the distributed AMF detector is given by

$$\Lambda_{\text{DAMF}} = \sum_{n=1}^N \max_{\Delta t_n, f_n} \tilde{q}_n(\Delta t_n, f_{d,n}). \quad (51)$$

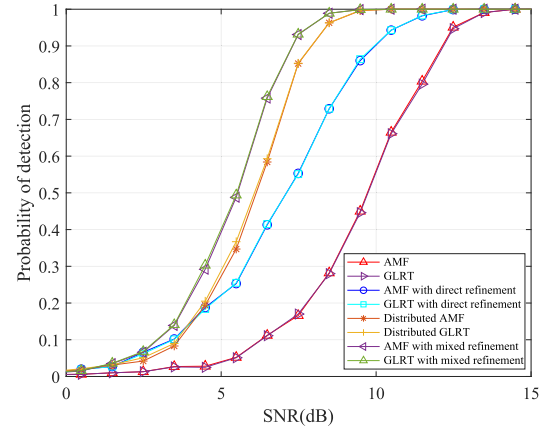
For comparison purposes, the CRLB for the target's position and velocity is also provided as a localization performance benchmark. The CRLB delineates the theoretical performance boundary of the radar network's parameter estimation and establishes a clear connection between the geometric configuration of the radar network and the estimation accuracy of the target's position and velocity. In practical applications, the CRLB can be computed based on the initial configuration of the radar network, serving as an important reference for assessing parameter estimation performance. Utilizing the CRLB, the geometric configuration of the



(a)



(b)



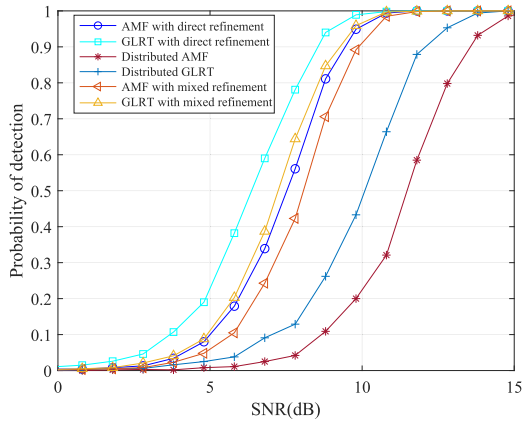
(c)

Fig. 6. Probability of target detection in the 2-D scenario with different offset errors. (a) $d = 2.23$. (b) $d = 3.35$. (c) $d = 5.01$.

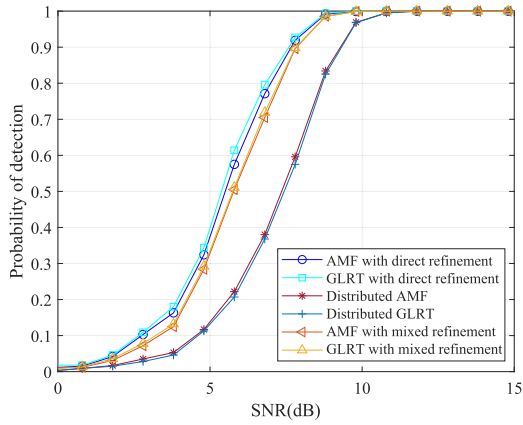
radar network can be reasonably arranged according to the expected accuracy for the target's position and velocity

$$\mathbf{C}_{p,v} = \left(\frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})^T}{\partial \boldsymbol{\varphi}} \right)^{-1} \quad (52)$$

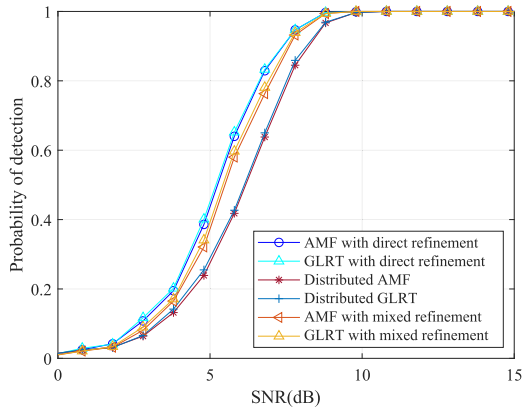
where $\mathbf{C}_{t,f}$ indicates CRLB in the TOA and Doppler of each node. The derivation and expression can be seen in Appendix C.



(a)



(b)



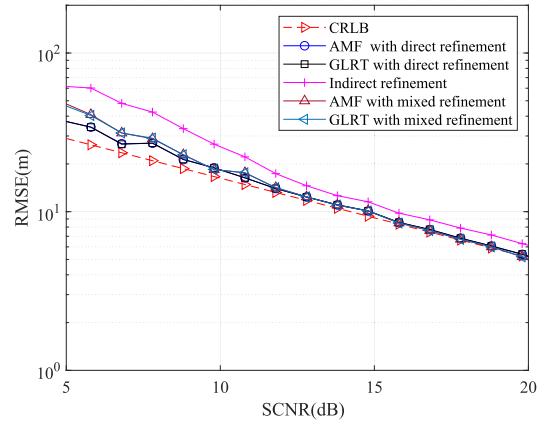
(c)

Fig. 7. Probability of target detection in the 2-D scenario with different sample support sizes of secondary data. (a) $K_1 = K_2 = \dots = K_N = 16$. (b) $K_1 = K_2 = \dots = K_N = 64$. (c) $K_1 = K_2 = \dots = K_N = 256$.

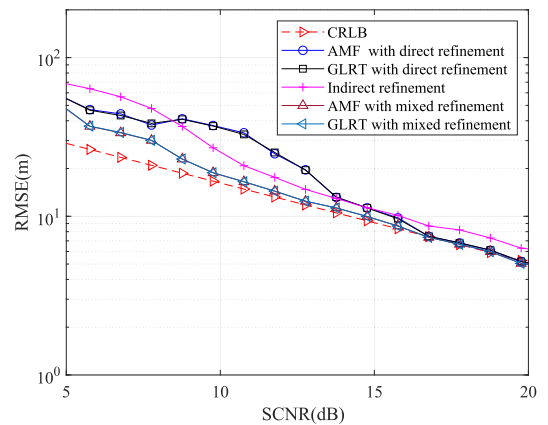
In order to describe the offset between the target's location and the initial point, a measurement based on the Mahalanobis distance is defined as

$$d = \sqrt{[\Delta \mathbf{p}^T, \Delta \mathbf{v}^T] \mathbf{C}_0^{-1} \begin{bmatrix} \Delta \mathbf{p} \\ \Delta \mathbf{v} \end{bmatrix}}. \quad (53)$$

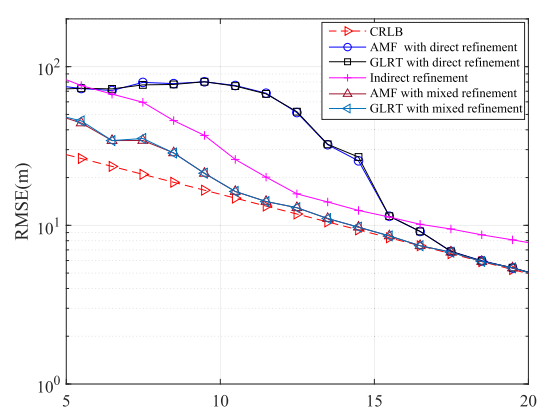
In the following, both 2-D and 3-D scenarios are designed to prove the effectiveness of the joint detection and localization



(a)



(b)



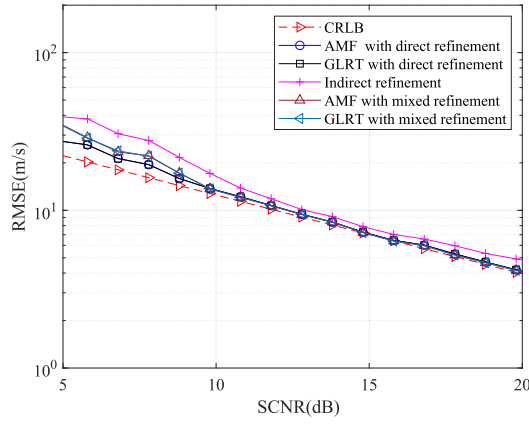
(c)

Fig. 8. Position estimation performance in the 2-D scenario with different offset errors. (a) $d = 2.23$. (b) $d = 3.35$. (c) $d = 5.01$.

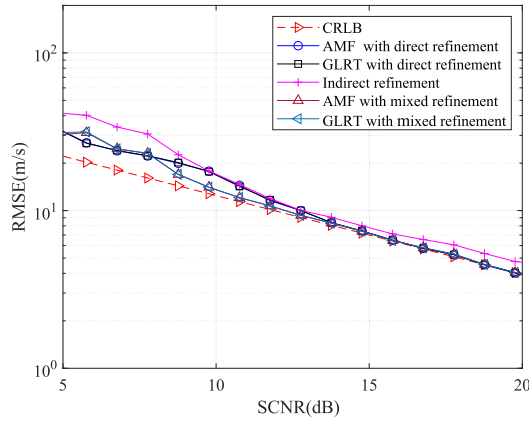
framework. In either a 2-D or 3-D case, the performance of the radar system pointing at the target can be analyzed.

A. Performance Analysis in the 2-D Scenario

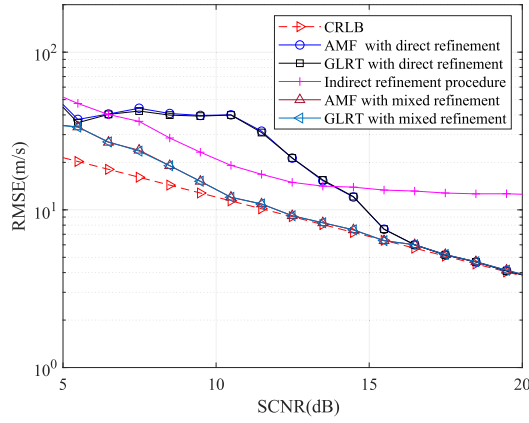
In the scenario, the radar network with five radar stations is considered. The values of bandwidth B_n are set as 0.9, 1, 1.2, 0.8, and 1.2 MHz, respectively. The values of carrier frequency are 0.33, 0.3, 0.36, 0.33, and 0.39 GHz, respectively.



(a)



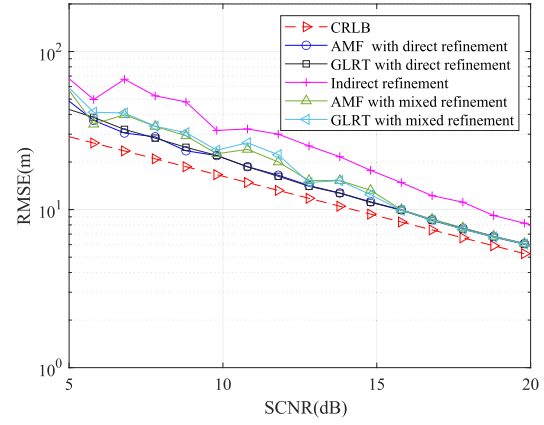
(b)



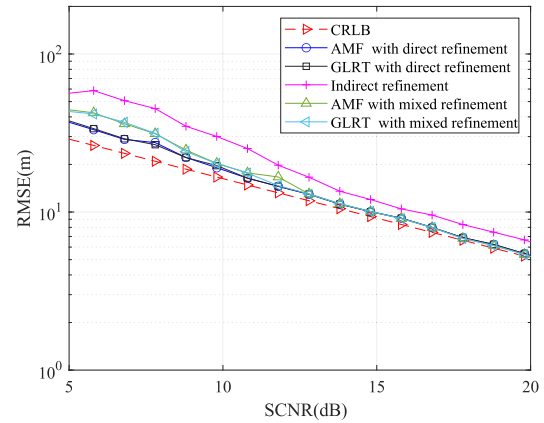
(c)

Fig. 9. Velocity estimation performance in the 2-D scenario with different offset errors. (a) $d = 2.23$. (b) $d = 3.35$. (c) $d = 5.01$.

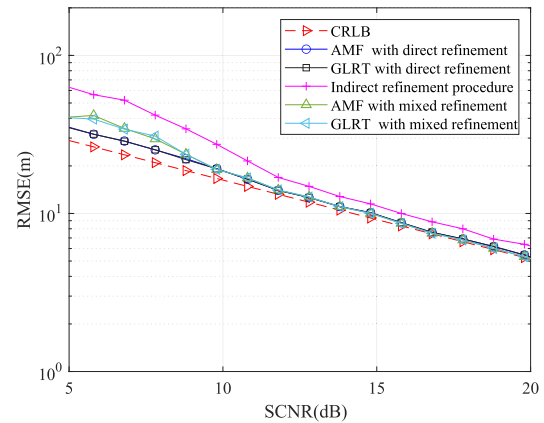
The transmitted signals' duration, pulse number, and PRI are $52 \mu s$, 4, and 1 ms, respectively. The signals are sampled at a very high rate of $20 \times B_n$, which is necessary to measure the peak location errors of the signal. An around placement is considered where we can obtain a good performance. As shown in Fig. 5, the placement of the radar nodes is located at (20,20), (−20, 20), (20, −20), (−20, −20), and (0,0) km, and the velocities of these radars are (0,10), (10,0), (−10, 0), (0, −10), and (0,0) m/s. In the common surveillance region



(a)



(b)

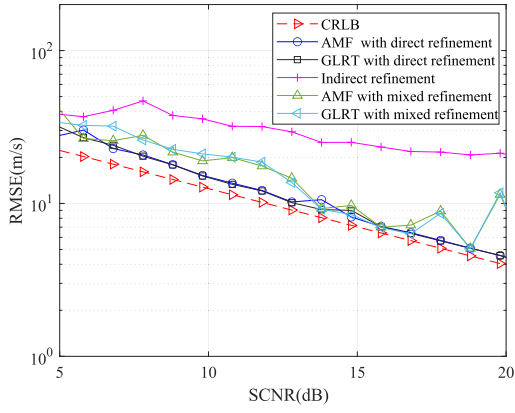


(c)

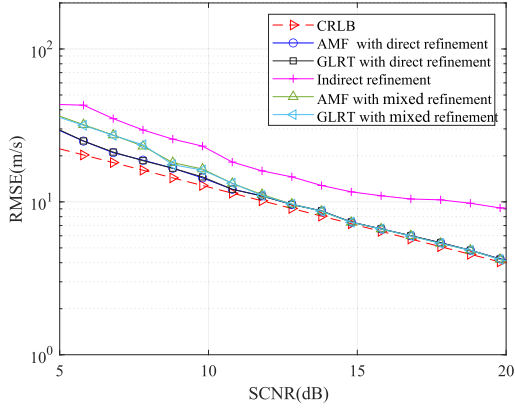
Fig. 10. Position estimation performance in the 2-D scenario with different sample support sizes of secondary data. (a) $K_1 = K_2 = \dots = K_N = 16$. (b) $K_1 = K_2 = \dots = K_N = 64$. (c) $K_1 = K_2 = \dots = K_N = 256$.

of the network, a target is placed in the (5,10) km and moves with (−100, −80) m/s. Considering the change of the radar cross section of the target, the SNR weights of different radar sites are set as 1.1, 1.2, 1, 0.8, and 0.9.

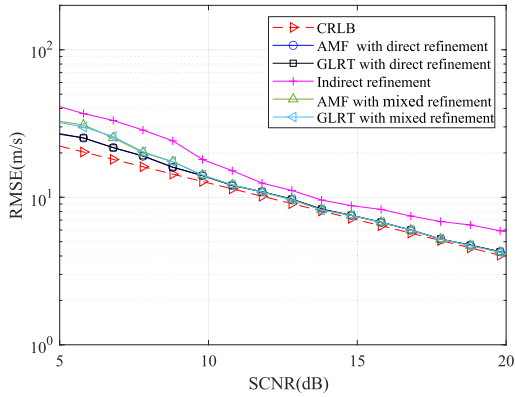
Fig. 6 shows the detection probability curves for the 2-D case with several target initial point, i.e., $d = 2.23$ ($\Delta \mathbf{p} = [60, 60]^T$, $\Delta \mathbf{v} = [20, 20]^T$), $d = 3.35$



(a)



(b)



(c)

Fig. 11. Velocity estimation performance in the 2-D scenario with different sample support sizes of secondary data. (a) $K_1 = K_2 = \dots = K_N = 16$. (b) $K_1 = K_2 = \dots = K_N = 64$. (c) $K_1 = K_2 = \dots = K_N = 256$.

($\Delta \mathbf{p} = [90, 90]^T$, $\Delta \mathbf{v} = [30, 30]^T$), and $d = 5.01$ ($\Delta \mathbf{p} = [120, 120]^T$, $\Delta \mathbf{v} = [60, 60]^T$). To study the upbound of the performance, a large secondary data set, i.e., $K_1 = K_2 = \dots = K_N = 1024$, is employed to train the sampling covariance matrix.

The experimental result highlights that the AMF and GLRT designs have very close performance in target detection in this case. Direct refinement, indirect refinement,

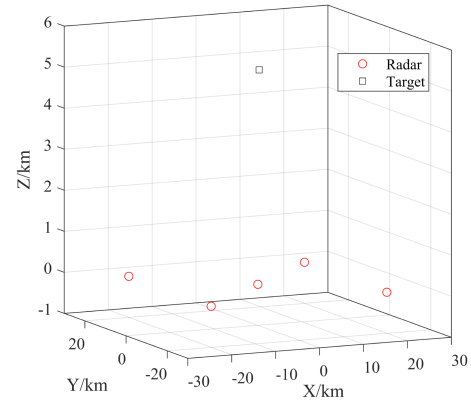
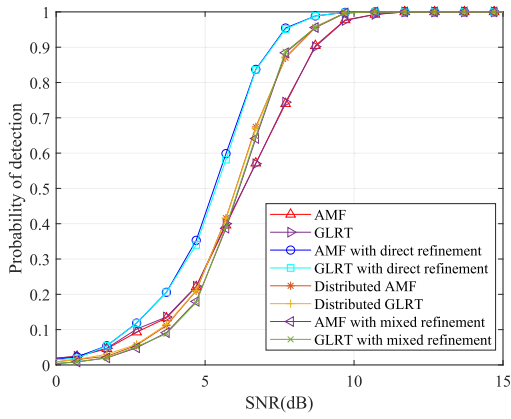


Fig. 12. Placement of radar nodes in the 3-D scenario.

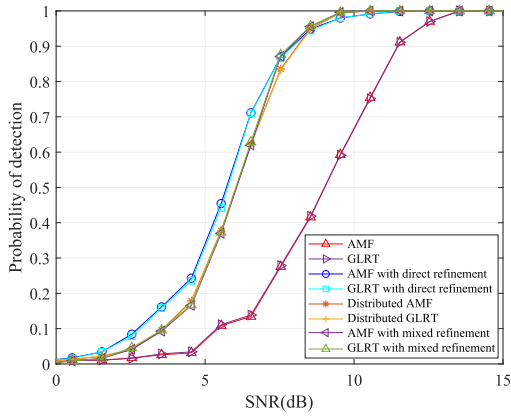
and mixed refinement achieve higher detection probability levels than the mismatched detectors (except for the case of $d = 0$). Specifically, Fig. 6(a) considers $d = 2.23$, where the starting point is satisfying (30) and (31) and the direct refinement has the best detection performance. As shown in Fig. 6(b) and (c), the direct methods have a performance degradation (about 2.5 dB in $d = 3.35$, 3.5 dB in $d = 5.01$) with respect to the case of $d = 2.23$, and the mixed-refinement-based detectors can keep a high level of performance. Thus, the direct method performs best when (30) and (31) are met, and the mixed method obtains higher performance than the direct one with a large offset.

In order to study detection performance with a constrained support size of secondary data, the probability of detection versus SCNR is displayed in Fig. 7. The sizes of secondary data are the same in all radars, where $K_1 = K_2 = \dots = K_N = K$, $K \in \{16, 64, 256\}$, and $d = 2.23$. Obviously, the detection performance improves with more secondary data. As shown in Fig. 7(a), when K keeps small, the GLRT-based detectors perform better than the AMF-based detectors. When K becomes large enough, the performance of the GLRT and AMF keeps a very close level. Comparing the indirect and direct methods, we find that the indirect method has a larger degradation with few secondary data, which shows the superiority of our proposed method.

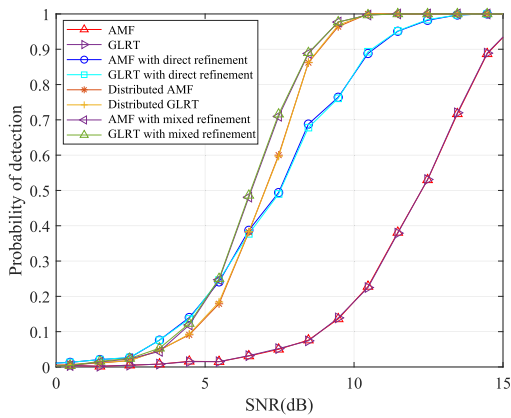
To assess the estimation capabilities of the proposed MLE in a 2-D case, the RMSE versus SCNR is displayed in Figs. 8 and 9 for different offsets between the initial starting point and target locations, i.e., $d = 2.23$, $d = 3.35$, and $d = 5.01$. Figs. 8 and 9 describe estimation performance of the position and velocity, respectively. The RMSE curves decrease with the SCNR, which is consistent with our expectation. Indeed, in the high-SCNR regime, the performance of the proposed method (direct refinement and mixed refinement) approaches the CRLB benchmark. When (30) and (31) are met, for example, $d = 2.23$, the direct refinement method behaves better than the indirect and mixed refinement methods. However, the mixed refinement method is the most effective for localization for a large offset. The results unambiguously demonstrate the effectiveness of the proposed estimators, including both AMF and GLRT. A significant amount of secondary data is used



(a)

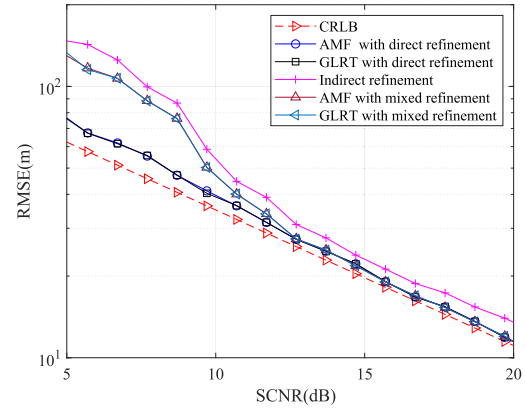


(b)

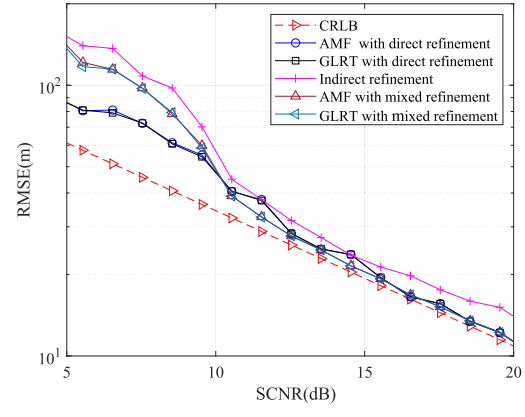


(c)

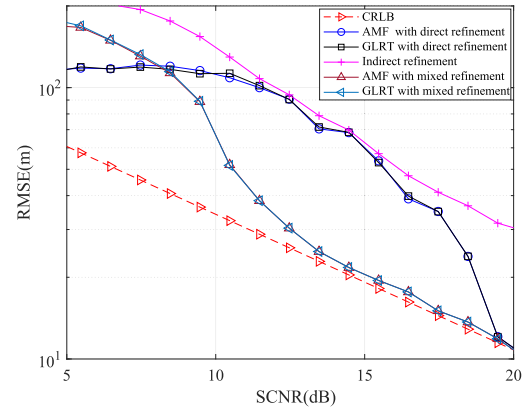
Fig. 13. Probability of target detection in the 3-D scenario with different offset errors. (a) $d = 2.23$. (b) $d = 3.35$. (c) $d = 5.01$.



(a)



(b)



(c)

Fig. 14. Position estimation performance in the 3-D scenario with different offset errors. (a) $d = 2.23$. (b) $d = 3.35$. (c) $d = 5.01$.

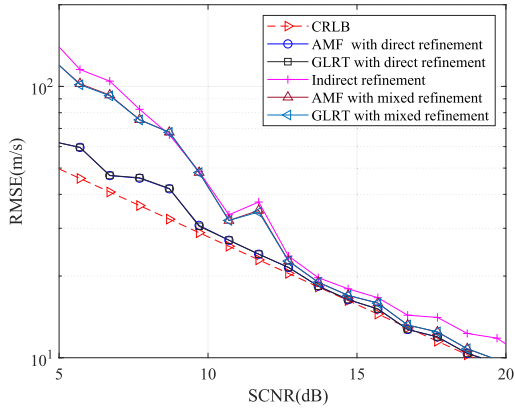
($K_1 = K_2 = \dots = K_N = 1024$); thus, the AMF design and the GLRT design have a very close estimation performance.

Figs. 10 and 11 display the relationship between the localization performance and the sizes of secondary data. The error offset keeps $d = 2.23$, and the sample support size is 16, 64, and 256. As expected, when K is large enough, the performance approaches the CRLB benchmark with SCNR increasing. A large performance loss occurred in estimation when K becomes small, especially velocity estimation, as shown in Fig. 11(a). Besides, the mixed method has an

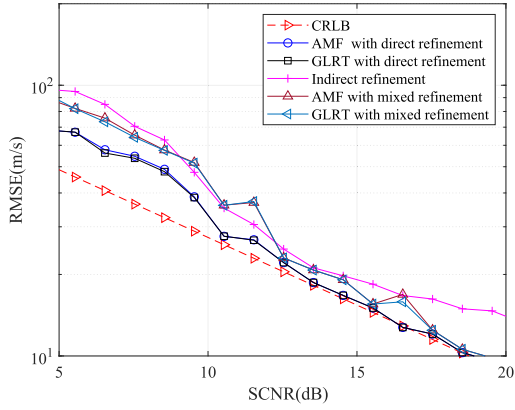
unstable performance because of the large error of indirect estimates.

B. Performance Analysis in the Actual 3-D Space

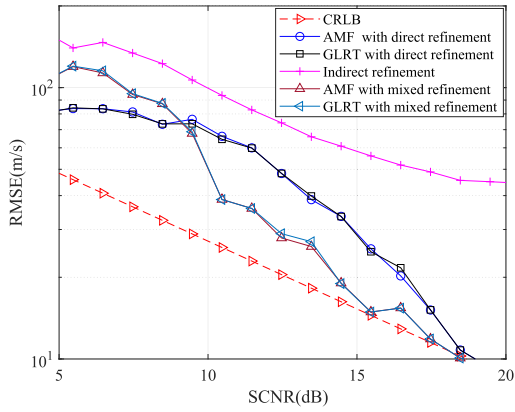
In this section, we consider the more actual 3-D scenario to further prove our algorithms' effectiveness. The parameters of the radars are the same as in the 2-D case. The deployment of the radar network, which involves five radars, is illustrated in Fig. 12. The radar nodes are located at (20,20,0), (−20, 20, 0), (20, −20, 0), (−20, −20, 0), and (0, 0, 0) km,



(a)



(b)

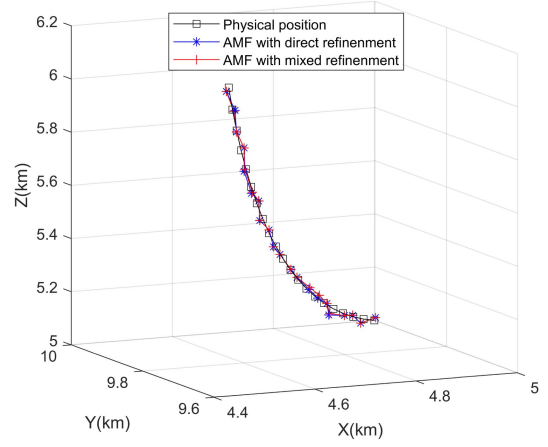


(c)

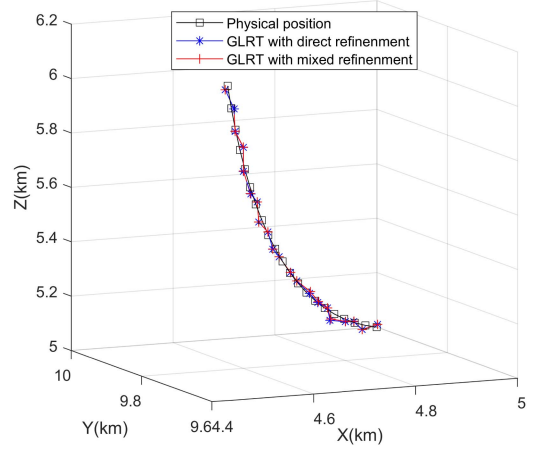
Fig. 15. Velocity estimation performance in the 3-D scenario with different offset errors. (a) $d = 2.23$. (b) $d = 3.35$. (c) $d = 5.01$.

and the velocities of these radars are $(0, 10, 10)$, $(10, 0, 0)$, $(-10, 0, 0)$, $(0, -10, 0)$, and $(0, 0, 0)$ m/s. The target is located at $(5, 10, 5)$ km and moves with $(-100, -80, 60)$ m/s. This is a typical scenario for an air search. The target is located in the common surveillance region of the radar network.

Figs. 13–15 present performance both for the 3-D scenarios with different error offset, where $d = 2.23$ ($\Delta \mathbf{p} = [60, 60, 60]^T$, $\Delta \mathbf{v} = [20, 20, 20]^T$), $d = 3.35$ ($\Delta \mathbf{p} = [90, 90, 90]^T$, $\Delta \mathbf{v} = [30, 30, 30]^T$), and $d = 5.01$



(a)



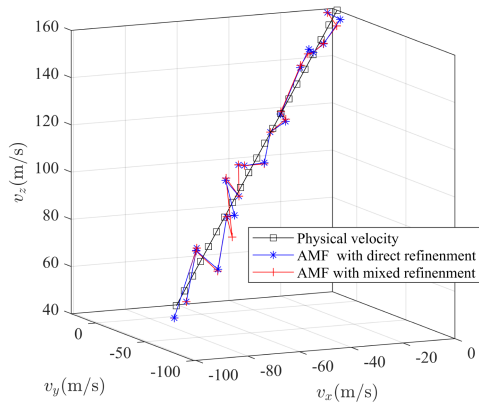
(b)

Fig. 16. Comparison of position estimates and physical plots. (a) Position estimates obtained by the AMF. (b) Position estimates obtained by GLRT.

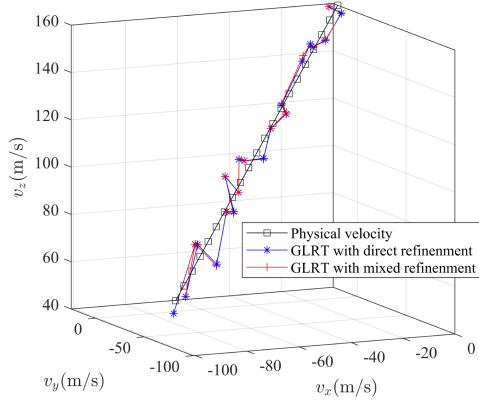
($\Delta \mathbf{p} = [120, 120, 120]^T$, $\Delta \mathbf{v} = [60, 60, 60]^T$). The size of secondary data is $K_1 = K_2 = \dots = K_N = 1024$.

Fig. 13 displays detection performance curves versus SCNR, where a comparison among the mismatch detectors, distributed detectors, and the proposed detectors is presented (both AMF and GLRT). The proposed adaptive detectors outperform the mismatch detectors and the distributed detectors, and with the increase of d , their performance is better and better than that of the mismatch detectors. The conclusion of the 2-D case is also suitable for this scenario. However, there is a performance loss of detection in this case with respect to the 2-D case. Again, the mixed refinement method obtains optimal performance with a large error offset, and the direct refinement performs best when the offset meets (30) and (31).

Figs. 14 and 15 show the position and velocity estimation performance versus SCNR, respectively. The performance curves show that the direct refinement method outperforms the indirect method in the condition of (30) and (31). In addition, it is worth noting that as the SCNR



(a)



(b)

Fig. 17. Comparison of velocity estimates and physical plots. (a) Velocity estimates obtained by the AMF. (b) Velocity estimates obtained by GLRT.

increases, the RMSE of all the estimators tends to decrease. Regardless of the setup, performance levels comparable to the CRLB benchmark are attained in the high-SCNR regime.

To further prove the effectiveness, our algorithms are applied to continuous observation of a moving target. The target moves with uniform acceleration. The initial start point is given, i.e., $d = 2.23$, which is used for the first observation. When the target needs to be reconfirmed, the localization results of the last observation are used as the initial starting point. Figs. 16 and 17 show results of continuous observation when the SCNR = 10 dB. The estimates around the true value show that our proposed algorithms are useful in the case of continuous observation.

V. CONCLUSION

This article addresses the complex task of simultaneous adaptive detection, position determination, and velocity estimation with a long-baseline radar network in 2-D and 3-D scenarios. Signal processing architectures have been proposed based on one-step GLRT and AMF, where two optimization strategies are introduced to calculate the precise estimates of the target's position and velocity under H_1

hypothesis. The first is the direct refinement method, where the iteration is implemented based on the steepest ascent initialized by the given rough point. In addition, the mixed refinement method is proposed for robustness with a large error offset, which can be regarded as the improved version of the direct method initialized by the outputs of the indirect iteration. Comparative experiments use the “detection then localization” methods available in the open literature. The results confirm the superior performance and effectiveness of the proposed adaptive detector. This method can be applied to practical engineering.

APPENDIX A

ML ESTIMATION OF α_n AND DERIVATION OF (20)

In this appendix, the decision rule (20) is deduced. The denominator of (19) can be the maximization of the following formula:

$$-\prod_{n=1}^N \left(-2 \sum_{l=1}^{L_n} \Re \{ \alpha_n^* s_{n,l} d_n^\dagger(p, v) S_n^{-1} d_n(p, v) \} + \sum_{l=1}^{L_n} |\alpha_n s_{n,l}(p)|^2 d_n^\dagger(p, v) S_n^{-1} d_n(p, v) + \sum_{l=1}^{L_n} z_{l,k}^\dagger S_n^{-1} z_{l,k} + 1 \right)^{K_n + L_n}. \quad (54)$$

For any p and v , (54) is maximized using $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_N$. Because signals received by different radars are independent, the complex amplitudes can be estimated individually. Note that the term $\sum_{l=1}^{L_n} z_{l,k}^\dagger S_n^{-1} z_{l,k} + 1$ is independent of α_n . Thus, for calculating $\hat{\alpha}_n$, the likelihood function can be further expressed as

$$\begin{aligned} \ell(Z_n | \alpha_n) &= 2 \sum_{l=1}^{L_n} \Re \{ \alpha_n^* s_{n,l} d_n^\dagger(p, v) S_n^{-1} d_n(p, v) \} \\ &\quad - \sum_{l=1}^{L_n} |\alpha_n s_{n,l}(p)|^2 d_n^\dagger(p, v) S_n^{-1} d_n(p, v). \end{aligned} \quad (55)$$

The likelihood function (55) is a scalar function in terms of real and imaginary parts of α_n ($\alpha_n = \alpha_n^R + j\alpha_n^I$). Taking the partial derivatives for the real part α_n^R and the imaginary part α_n^I , respectively, $\hat{\alpha}_n$ can be obtained by solving

$$\frac{\partial}{\partial \alpha_n^R} \ell(Z_n | \alpha_n^R) = 0 \quad (56)$$

$$\frac{\partial}{\partial \alpha_n^I} \ell(Z_n | \alpha_n^I) = 0. \quad (57)$$

By solving (56) and (57), the real and imaginary parts of α_n are given as

$$\hat{\alpha}_n^R = \frac{\sum_{l=1}^{L_n} \Re \{ s_{n,l}(p) d_n^\dagger(p, v) S_n^{-1} z_{n,l} \}}{\sum_{l=1}^{L_n} s_{n,l}(p) d_n^\dagger(p, v) S_n^{-1} d_n(p, v)} \quad (58)$$

$$\hat{\alpha}_n^I = \frac{\sum_{l=1}^{L_n} j \Im \{ s_{n,l}(p) d_n^\dagger(p, v) S_n^{-1} z_{n,l} \}}{\sum_{l=1}^{L_n} s_{n,l}(p) d_n^\dagger(p, v) S_n^{-1} d_n(p, v)}. \quad (59)$$

Combining (58) and (59), we have

$$\hat{\alpha}_n = \frac{\sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{z}_{n,l}}{\sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{d}_n^\dagger(\mathbf{p}, \mathbf{v}) \mathbf{S}_n^{-1} \mathbf{d}_n(\mathbf{p}, \mathbf{v})}. \quad (60)$$

Substituting (60) into (19), the statistic can be given by

$$\max_{\mathbf{p}, \mathbf{v}} \prod_{n=1}^N \left(1 - \frac{q_n(\mathbf{p}, \mathbf{v})}{p_n}\right)^{-(K_n+L_n)} \underset{H_0}{\geq} \xi. \quad (61)$$

APPENDIX B

PARTIAL DERIVATIVES OF $\partial q'_n(\Delta t_n, f_{d,n})$ FOR Δt_n AND $f_{d,n}$

For convenience, we define the “whitened” quantities

$$\begin{aligned} \tilde{\chi} &= \sum_{l=1}^{L_n} s_{n,l}(\mathbf{p}) \mathbf{S}_n^{-1/2} \mathbf{z}_{n,l} \\ &= \sum_{l=1}^{L_n} s'_{n,l}(\Delta t_n) \mathbf{S}_n^{-1/2} \mathbf{z}_{n,l} \end{aligned} \quad (62)$$

$$\begin{aligned} \tilde{\mathbf{d}}_n &= \mathbf{S}_n^{-1/2} \mathbf{d}_n(\mathbf{p}, \mathbf{v}) \\ &= \mathbf{S}_n^{-1/2} \mathbf{d}'_n(\Delta t_n, f_{d,n}). \end{aligned} \quad (63)$$

Given that $\|\mathbf{s}_n(\mathbf{p})\| = \sum_{l=1}^{L_n} |s_{n,l}(\mathbf{p})|^2 = \|\mathbf{s}_n\|$, we recast (21)

$$q'_n(\Delta t_n, f_{d,n}) = q_n(\mathbf{p}, \mathbf{v}) = \frac{|\tilde{\mathbf{d}}_n^\dagger \tilde{\chi}|^2}{\|\mathbf{s}_n\|^2 \|\tilde{\mathbf{d}}_n\|_2^2}. \quad (64)$$

The partial derivative of the function $q'_n(\Delta t_n, f_{d,n})$ for Δt_n is given as

$$\frac{\partial q'_n(\Delta t_n, f_{d,n})}{\partial \Delta t_n} = \frac{2\Re \left\{ \frac{\partial \tilde{\chi}}{\partial \Delta t_n} \tilde{\mathbf{d}}_n^\dagger \mathbf{S}_n^{-1} \tilde{\mathbf{d}}_n^\dagger \tilde{\chi} \right\}}{\|\mathbf{s}_n\|^2 \tilde{\mathbf{d}}_n^\dagger \tilde{\mathbf{d}}_n} \quad (65)$$

where $\frac{\partial \tilde{\chi}}{\partial \Delta t_n}$ can be obtained as

$$\frac{\partial \tilde{\chi}}{\partial \Delta t_n} = \sum_{l=1}^{L_n} \mathbf{S}_n^{-1/2} \mathbf{z}_{n,l} \frac{\partial s'_{n,l}(\Delta t_n)}{\partial \Delta t_n}. \quad (66)$$

The partial derivative for $f_{d,n}$ is obtained by

$$\begin{aligned} \frac{\partial q'_n(\Delta t_n, f_{d,n})}{\partial f_{d,n}} &= \frac{2\Re \left\{ \frac{\partial \tilde{\mathbf{d}}_n}{\partial f_{d,n}} \tilde{\chi} \tilde{\chi}^\dagger \tilde{\mathbf{d}}_n \right\} \tilde{\mathbf{d}}_n^\dagger \tilde{\mathbf{d}}_n}{\|\mathbf{s}_n\|^2 \left[\tilde{\mathbf{d}}_n^\dagger \tilde{\mathbf{d}}_n \right]} \\ &\quad - \frac{2\Re \left\{ \frac{\partial \tilde{\mathbf{d}}_n}{\partial f_{d,n}} \tilde{\mathbf{d}}_n^\dagger \right\} \tilde{\mathbf{d}}_n^\dagger \tilde{\chi} \tilde{\chi}^\dagger \tilde{\mathbf{d}}_n}{\|\mathbf{s}_n\|^2 \left[\tilde{\mathbf{d}}_n^\dagger \tilde{\mathbf{d}}_n \right]^2} \end{aligned} \quad (67)$$

where the term $\frac{\partial \tilde{\mathbf{d}}_n}{\partial f_{d,n}} \in \mathbb{C}^{M_n \times 1}$ is given by

$$\frac{\partial \tilde{\mathbf{d}}_n}{\partial f_{d,n}} = j2\pi T_{r,n} \begin{bmatrix} 0 \\ e^{j2\pi f_{d,n} T_{r,n}} \\ \vdots \\ (M_n - 1)e^{j2\pi f_{d,n} (M_n - 1) T_{r,n}} \end{bmatrix}. \quad (68)$$

The terms $\frac{\partial \Delta t_n}{\partial \mathbf{p}} \in \mathbb{R}^{3 \times 1}$ and $\frac{\partial f_{d,n}}{\partial \mathbf{p}} \in \mathbb{R}^{3 \times 1}$ in (28) and (46) are, respectively, given as

$$\frac{\partial \Delta t_n}{\partial \mathbf{p}} = \frac{2(\mathbf{p} - \mathbf{p}_{R,n})}{c \|\mathbf{p} - \mathbf{p}_{R,n}\|} \quad (69)$$

$$\begin{aligned} \frac{\partial f_{d,n}}{\partial \mathbf{p}} &= \frac{2(\mathbf{v} - \mathbf{v}_{R,n})}{\lambda_n \|\mathbf{p} - \mathbf{p}_{R,n}\|} \\ &\quad + \frac{2(\mathbf{p} - \mathbf{p}_{R,n})^\top (\mathbf{v} - \mathbf{v}_{R,n})(\mathbf{p} - \mathbf{p}_{R,n})}{\lambda_n \|\mathbf{p} - \mathbf{p}_{R,n}\|^3}. \end{aligned} \quad (70)$$

The partial derivative $\frac{\partial f_{d,n}}{\partial \mathbf{v}} \in \mathbb{R}^{3 \times 1}$ in (29) and (47) is

$$\frac{\partial f_{d,n}}{\partial \mathbf{v}} = \frac{2(\mathbf{p} - \mathbf{p}_{R,n})}{\lambda_n \|\mathbf{p} - \mathbf{p}_{R,n}\|}. \quad (71)$$

Because the term Δt_n is not the function of $\mathbf{v}$, we have $\frac{\partial \Delta t_n}{\partial \mathbf{v}} = \mathbf{0}$.

APPENDIX C

CRLB CALCULATION

In the following, the CRLB evaluation for the estimation of $\mathbf{p}$ and $\mathbf{v}$ is conducted, assuming the independence of each of the radar nodes. From (8), we can rewrite the signal model as a vector form

$$\mathbf{y}_n = \alpha_n \mathbf{s}'_n(\Delta t_n) \otimes \mathbf{d}'_n(f_{d,n}) + \mathbf{w}_n \quad (72)$$

where $\mathbf{s}'_n(\Delta t_n) = [s'_{n,1}(\Delta t_n), s'_{n,2}(\Delta t_n), \dots, s'_{n,L_n}(\Delta t_n)]$. $\mathbf{w}_n$ is noise plus clutter with zero-mean and the covariance matrix $\tilde{\mathbf{C}}_n = \mathbf{C}_n \otimes \mathbf{I}_{L_n} \in \mathbb{C}^{M_n L_n \times M_n L_n}$. Then, the CRLB of the model (72) is given in [57], which enables us to concentrate it onto the parameters of interest Δt_n and $f_{d,n}$. The expression is given as

$$\begin{aligned} \text{CRLB}_{\Delta t_n, \Delta t_n} &= \frac{1}{2|\alpha_n|^2 \mathbf{d}'^\dagger(f_{d,n}) \mathbf{C}_n^{-1} \mathbf{d}'(f_{d,n})} \\ &\quad \times \Re \left\{ \frac{\partial \mathbf{s}'_n(\Delta t_n)^\dagger}{\partial \Delta t_n} \mathbf{\Pi}_{s,n}^\perp \frac{\partial \mathbf{s}'_n(\Delta t_n)}{\partial \Delta t_n} \right\}^{-1} \end{aligned} \quad (73)$$

$$\text{CRLB}_{\Delta t_n, f_{d,n}} = 0 \quad (74)$$

$$\begin{aligned} \text{CRLB}_{f_{d,n}, f_{d,n}} &= \frac{1}{2|\alpha_n|^2 \mathbf{s}'^\dagger_n(\Delta t_n) \mathbf{s}'_n(\Delta t_n)} \\ &\quad \times \Re \left\{ \frac{\partial \mathbf{d}'_n(f_{d,n})^\dagger}{\partial f_{d,n}} \mathbf{C}^{-1/2} \mathbf{\Pi}_{d,n}^\perp \mathbf{C}^{-1/2} \frac{\partial \mathbf{d}'_n(f_{d,n})}{\partial f_{d,n}} \right\}^{-1} \end{aligned} \quad (75)$$

where

$$\mathbf{\Pi}_{s,n}^\perp = \mathbf{I}_{L_n} - \frac{\mathbf{s}'_n(\Delta t_n) \mathbf{s}'^\dagger_n(\Delta t_n)}{\mathbf{s}'^\dagger_n(\Delta t_n) \mathbf{s}'_n(\Delta t_n)} \quad (76)$$

$$\mathbf{\Pi}_{d,n}^\perp = \mathbf{I}_M - \frac{\mathbf{C}_n^{-1/2} \mathbf{d}'_n(f_{d,n}) \mathbf{d}'^\dagger_n(f_{d,n}) \mathbf{C}_n^{-1/2}}{\mathbf{d}'^\dagger_n(f_{d,n}) \mathbf{C}_n^{-1} \mathbf{d}'_n(f_{d,n})}. \quad (77)$$

Combining CRLB of all radar nodes, we obtain

$$\mathbf{C}_{t,f} = \text{diag}(\text{CRLB}_{\Delta t_1, \Delta t_1}, \dots, \text{CRLB}_{\Delta t_N, \Delta t_N}, \text{CRLB}_{f_{d,1}, f_{d,1}}, \dots, \text{CRLB}_{f_{d,N}, f_{d,N}}). \quad (78)$$

For convenience, we define

$$\boldsymbol{\varphi} = \begin{bmatrix} \mathbf{p} \\ \mathbf{v} \end{bmatrix} \quad (79)$$

$$\boldsymbol{\varphi} = [t_1(\mathbf{p}), \dots, t_N(\mathbf{p}), \quad (80)$$

$$f_{d,1}(\mathbf{p}, \mathbf{v}), \dots, f_{d,N}(\mathbf{p}, \mathbf{v})]^T \quad (81)$$

where $\boldsymbol{\varphi} = \mathbf{g}(\boldsymbol{\theta})$. $\hat{\boldsymbol{\varphi}}$ and $\hat{\boldsymbol{\theta}}$ are denotes the estimates of $\boldsymbol{\varphi}$ and $\boldsymbol{\theta}$, respectively. We have $\mathbb{E}\{\hat{\boldsymbol{\theta}}\} = \boldsymbol{\theta} = \mathbf{g}(\boldsymbol{\varphi})$.

According to [58], we have an identical equation

$$\int (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \frac{\partial \ln f(\mathbf{Z}|\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T f(\mathbf{Z}|\boldsymbol{\varphi}) d\mathbf{y} = \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \quad (82)$$

where $f(\mathbf{Z}|\boldsymbol{\varphi})$ is the conditional PDF under H_1 . $\frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \in \mathbb{R}^{6 \times 6}$ is defined as

$$\frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} = \begin{bmatrix} \frac{\partial \Delta t_1}{\partial \mathbf{p}}^T & \frac{\partial \Delta t_1}{\partial \mathbf{p}}^T \\ \vdots & \vdots \\ \frac{\partial \Delta t_N}{\partial \mathbf{p}}^T & \frac{\partial \Delta t_N}{\partial \mathbf{p}}^T \\ \mathbf{0}_{N \times 3} & \frac{\partial f_{d,1}}{\partial \mathbf{v}}^T \\ & \vdots \\ & \frac{\partial f_{d,N}}{\partial \mathbf{v}}^T \end{bmatrix}. \quad (83)$$

For obtaining the CRLB of $\boldsymbol{\varphi}$, an auxiliary equation is given as

$$\int \mathbf{a}^T (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \frac{\partial \ln f(\mathbf{Z}|\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T f(\mathbf{Z}|\boldsymbol{\varphi}) \mathbf{b}^T d\mathbf{y} = \mathbf{a}^T \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{b}^T \quad (84)$$

where $\mathbf{a} \in \mathbb{R}^{2N \times 1}$ and $\mathbf{b} \in \mathbb{C}^{6 \times 1}$ are arbitrary constant vectors. Depending on the Cauchy-Schwarz inequality, (84) can be rewritten as

$$\left(\mathbf{a}^T \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{b}^T \right)^2 \leq \int \mathbf{a}^T (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T \mathbf{a} f(\mathbf{Z}|\boldsymbol{\varphi}) d\mathbf{t} \times \int \mathbf{b}^T \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \mathbf{b} f(\mathbf{Z}|\boldsymbol{\varphi}) d\mathbf{t}. \quad (85)$$

Note that

$$\mathbb{E} \left[\frac{\partial \ln f(\mathbf{Z}|\boldsymbol{\varphi})}{\partial \varphi_a} \frac{\partial \ln f(\mathbf{Z}|\boldsymbol{\varphi})}{\partial \varphi_b} \right] = -\mathbb{E} \left[\frac{\partial^2 \ln f(\mathbf{Z}|\boldsymbol{\varphi})}{\partial \varphi_a \partial \varphi_b} \right] = [\mathbf{F}_{p,v}(\boldsymbol{\varphi})]_{a,b} \quad (86)$$

where $\mathbf{F}_{p,v}(\boldsymbol{\varphi})$ is the Fisher information matrix for estimating $\mathbf{p}$ and $\mathbf{v}$.

Since $\mathbf{a}$ is arbitrary, let

$$\mathbf{a} = \mathbf{C}_{t,f} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \mathbf{b} \quad (87)$$

to yield

$$\left(\mathbf{b}^T \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \mathbf{b} \right)^2 \leq \mathbf{b}^T \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \mathbf{b} \mathbf{b}^T \mathbf{F}_{p,v}(\boldsymbol{\varphi}) \mathbf{b}. \quad (88)$$

Because $\mathbf{C}_{t,f}$ is positive definite, so is $\mathbf{C}_{t,f}^{-1}$. The term $\frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T$ is at least positive semidefinite. Thus, let both sides of the inequality be divided the following non-negative term:

$$\mathbf{b}^T \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \mathbf{b}. \quad (89)$$

Then, we have

$$\mathbf{b}^T \left(\mathbf{F}_{p,v}(\boldsymbol{\varphi}) - \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \right) \mathbf{b} \geq \mathbf{0}. \quad (90)$$

Finally, the CRLB of $\mathbf{p}$ and $\mathbf{v}$ can be obtained by

$$\mathbf{C}_{p,v} = \left(\frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}} \mathbf{C}_{t,f}^{-1} \frac{\partial \mathbf{g}(\boldsymbol{\varphi})}{\partial \boldsymbol{\varphi}}^T \right)^{-1}. \quad (91)$$

REFERENCES

- [1] S. H. Javadi and A. Farina, "Radar networks: A review of features and challenges," *Inf. Fusion*, vol. 61, pp. 48–55, 2020.
- [2] K. V. Shanbhag, D. Deb, and M. Kulkarni, "MIMO radar with spatial-frequency diversity for improved detection performance," in *Proc. Int. Conf. Commun. Control Comput. Technol.*, 2010, pp. 66–70.
- [3] Q. He and R. S. Blum, "Diversity gain for MIMO Neyman-Pearson signal detection," *IEEE Trans. Signal Process.*, vol. 59, no. 3, pp. 869–881, Mar. 2011.
- [4] H. Liu, S. Zhou, H. Su, and Y. Yu, "Detection performance of spatial-frequency diversity MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 4, pp. 3137–3155, Oct. 2014.
- [5] Z. Yu, J. Li, Q. Guo, and J. Ding, "Efficient direct target localization for distributed MIMO radar with expectation propagation and belief propagation," *IEEE Trans. Signal Process.*, vol. 69, pp. 4055–4068, 2021.
- [6] M. Einemo and H. C. So, "Weighted least squares algorithm for target localization in distributed MIMO radar," *Signal Process.*, vol. 115, pp. 144–150, 2015.
- [7] J. Guo and H. Tao, "Cramer-Rao lower bounds of target positioning estimate in netted radar system," *Digit. Signal Process.*, vol. 118, 2021, Art. no. 103222.
- [8] S. Zhang, Y. Zhou, L. Zhang, Q. Zhang, and L. Du, "Target detection for multistatic radar in the presence of deception jamming," *IEEE Sens. J.*, vol. 21, no. 6, pp. 8130–8141, Mar. 2021.
- [9] Q. Li et al., "Hermitian distance-based method to discriminate physical targets and active false targets in a distributed multiple-radar architecture," *IEEE Sens. J.*, vol. 19, no. 22, pp. 10432–10442, Nov. 2019.
- [10] L. Lan, G. Liao, J. Xu, Y. Zhang, and S. Zhu, "Mainlobe deceptive jammer suppression using element-pulse coding with MIMO radar," *Signal Process.*, vol. 182, 2021, Art. no. 107955.
- [11] V. S. Chernyak, *Fundamentals of Multisite Radar Systems: Multistatic Radars and Multiradar Systems*. London, U.K.: Gordon & Breach, 1998.
- [12] S. G. Dontamsetti and R. V. R. Kumar, "A distributed MIMO radar with joint optimal transmit and receive signal combining," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 1, pp. 623–635, Feb. 2021.
- [13] X. Xiao, S. Li, S. Peng, X. Xue, X. Zheng, and B. Zhou, "Microwave photonic wideband distributed coherent aperture radar with high robustness to time synchronization error," *J. Lightw. Technol.*, vol. 39, no. 2, pp. 347–356, Jan. 2021.

- [14] M. Sadeghi, F. Behnia, R. Amiri, and A. Farina, "Target localization geometry gain in distributed MIMO radar," *IEEE Trans. Signal Process.*, vol. 69, pp. 1642–1652, 2021.
- [15] R. Hoffmann, N. Neuberger, and R. Vehmas, "Rx beamforming for long baseline multistatic radar networks," in *Proc. IEEE Radar Conf.*, 2021, vol. 21, pp. 1–6.
- [16] H. Godrich, A. M. Haimovich, and R. S. Blum, "Target localization accuracy gain in MIMO radar-based systems," *IEEE Trans. Inf. Theory*, vol. 56, no. 6, pp. 2783–2803, Jun. 2010.
- [17] Q. He, R. S. Blum, H. Godrich, and A. M. Haimovich, "Target velocity estimation and antenna placement for MIMO radar with widely separated antennas," *IEEE J. Sel. Topics Signal Process.*, vol. 4, no. 1, pp. 79–100, Feb. 2010.
- [18] Y. Zhao, Y. Zhao, and C. Zhao, "A novel algebraic solution for moving target localization in multi-transmitter multi-receiver passive radar," *Signal Process.*, vol. 143, pp. 303–310, 2018.
- [19] F. Zhang, Y. Sun, J. Zou, D. Zhang, and Q. Wan, "Closed-form localization method for moving target in passive multi-static radar network," *IEEE Sens. J.*, vol. 20, no. 2, pp. 980–990, Jan. 2020.
- [20] A. Noroozi, R. Amiri, M. Nayeibi, and A. Farina, "Efficient closed-form solution for moving target localization in MIMO radars with minimum number of antennas," *IEEE Trans. Signal Process.*, vol. 68, pp. 2545–2557, 2020.
- [21] A. Noroozi, M. Sebt, and A. Oveis, "Efficient weighted least squares estimator for moving target localization in distributed MIMO radar with location uncertainties," *IEEE Syst. J.*, vol. 13, no. 4, pp. 4454–4463, Dec. 2019.
- [22] R. Niu, P. K. Varshney, and Q. Cheng, "Distributed detection in a large wireless sensor network," *Inf. Fusion*, vol. 7, no. 4, pp. 380–394, 2006.
- [23] S. H. Javadi and A. Peiravi, "Fusion of weighted decisions in wireless sensor networks," *IET Wireless Sens. Syst.*, vol. 5, no. 2, pp. 97–105, 2015.
- [24] Q. He, N. H. Lehmann, R. S. Blum, and A. M. Haimovich, "MIMO radar moving target detection in homogeneous clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 46, no. 3, pp. 1290–1301, Jul. 2010.
- [25] R. Amiri, F. Behnia, and A. Noroozi, "Efficient joint moving target and antenna localization in distributed MIMO radars," *IEEE Trans. Wireless Commun.*, vol. 18, no. 9, pp. 4425–4435, Sep. 2019.
- [26] T. Sun, C. Dong, Y. Mao, and M. Liu, "Moving target localization in multiple-input multiple-output radar systems without the prior knowledge of measurement noise powers," *IEEE Commun. Lett.*, vol. 24, no. 9, pp. 1957–1960, Sep. 2020.
- [27] Y. Zhang and K. Ho, "Multistatic moving object localization by a moving transmitter of unknown location and offset," *IEEE Trans. Signal Process.*, vol. 68, pp. 4438–4453, 2020.
- [28] X. Wu, Y. Liu, X. Zhu, and L. Mo, "Efficient solutions for MIMO radar localization under unknown transmitter positions and offsets," *IEEE Trans. Wireless Commun.*, vol. 21, no. 1, pp. 505–518, Jan. 2022.
- [29] G. Wen, H. Song, F. Gao, Y. Liang, and L. Zhu, "Target localization in asynchronous distributed MIMO radar systems with a cooperative target," *IEEE Trans. Wireless Commun.*, vol. 21, no. 6, pp. 4098–4113, Jun. 2022.
- [30] T. Jia, H. Wang, X. Shen, Z. Jiang, and K. He, "Target localization based on structured total least squares with hybrid TDOA-AOA measurements," *Signal Process.*, vol. 143, pp. 211–221, 2018.
- [31] T. Jia, H. Wang, G. Wang, and K. C. Ho, "Localization using time-delay and Doppler shift by moving monostatic sensors," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 3, pp. 2560–2567, Jun. 2022.
- [32] Z. Shi, H. Wang, C. S. Leung, H. C. So, and M. EURASIP, "Robust MIMO radar target localization based on Lagrange programming neural network," *Signal Process.*, vol. 174, 2020, Art. no. 107574.
- [33] Y. Yang, H. Su, Q. Hu, S. Zhou, and J. Huang, "Spatial resolution cell based centralized target detection in multistatic radar," *Signal Process.*, vol. 152, pp. 238–246, 2018.
- [34] Z. Zhang, Y. Zhou, S. Zhang, S. Shang, T. You, and L. Zhang, "Elliptical spatial resolution cell-based centralized target detection in multistatic radar," *IEEE Sens. J.*, vol. 24, no. 9, pp. 15367–15377, May 2024.
- [35] O. Bar-Shalom and A. J. Weiss, "Direct positioning of stationary targets using MIMO radar," *Signal Process.*, vol. 91, no. 10, pp. 2345–2358, 2011.
- [36] W. Liu, J. Liu, H. Li, Q. Du, and Y.-L. Wang, "Multichannel signal detection based on Wald test in subspace interference and Gaussian noise," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 3, pp. 1370–1381, Jun. 2019.
- [37] W. Liu, J. Liu, Y. Gao, G. Wang, and Y.-L. Wang, "Multichannel signal detection in interference and noise when signal mismatch happens," *Signal Process.*, vol. 166, 2020, Art. no. 107268.
- [38] O. Besson, "Rao, Wald, and Gradient tests for adaptive detection of Swerling I targets," *IEEE Trans. Signal Process.*, vol. 71, pp. 3043–3052, 2023.
- [39] W. Liu, J. Liu, Y. Gao, G. Wang, and Y.-L. Wang, "Multichannel adaptive signal detection: Basic theory and literature review," *Sci. China Inf. Sci.*, vol. 65, 2022, Art. no. 121301.
- [40] S. Zhou, "Signal fusion-based target detection algorithm for spatial diversity radar," *IET Radar, Sonar Navigat.*, vol. 5, no. 3, pp. 204–214, Mar. 2011.
- [41] Y. Yang, H. Su, Q. Hu, S. Zhou, and J. Huang, "Centralized adaptive CFAR detection with registration errors in multistatic radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 5, pp. 2370–2382, Oct. 2018.
- [42] Q. Hu, H. Su, S. Zhou, Z. Liu, and J. Liu, "Target detection in distributed MIMO radar with registration errors," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 52, no. 1, pp. 438–450, Feb. 2016.
- [43] J. Bosse, O. Krasnov, and A. Yarovoy, "Direct target localization with an active radar network," *Signal Process.*, vol. 125, pp. 21–35, 2016.
- [44] J. Bosse, O. Krasnov, and A. Yarovoy, "Direct target localization and deghosting in active radar network," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 4, pp. 3139–3150, Oct. 2015.
- [45] G. V. Moustakides, G. H. Jajamovich, A. Tajer, and X. Wang, "Joint detection and estimation: Optimum tests and applications," *IEEE Trans. Inf. Theory*, vol. 58, no. 7, pp. 4215–4229, Jul. 2012.
- [46] D. Orlando and G. Ricci, "Adaptive radar detection and localization of a point-like target," *IEEE Trans. Signal Process.*, vol. 59, no. 9, pp. 4086–4096, Sep. 2011.
- [47] M. Xing, J. Su, G. Wang, and Z. Bao, "New parameter estimation and detection algorithm for high speed small target," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 1, pp. 214–224, Jan. 2011.
- [48] A. Aubry, A. D. Maio, S. Marano, and M. Rosamilia, "Single-pulse simultaneous target detection and angle estimation in a multi-channel phased array radar," *IEEE Trans. Signal Process.*, vol. 68, pp. 6649–6664, 2020.
- [49] F. Santi, F. Pieralice, and D. Pastina, "Joint detection and localization of vessels at sea with a GNSS-based multistatic radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5894–5913, Aug. 2019.
- [50] W. Yi, T. Zhou, Y. Ai, and R. S. Blum, "Suboptimal low complexity joint multi-target detection and localization for non-coherent MIMO radar with widely separated antennas," *IEEE Trans. Signal Process.*, vol. 68, pp. 901–916, 2020.
- [51] E. Kelly, "An adaptive detection algorithm," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-22, no. 2, pp. 115–127, Mar. 1986.
- [52] A. Aubry, A. De Maio, G. Foglia, C. Hao, and D. Orlando, "A radar detector with enhanced range estimation capabilities for partially homogeneous environment," *IET Radar, Sonar Navigat.*, vol. 8, no. 9, pp. 1018–1025, 2014.
- [53] F. Robey, D. Fuhrmann, E. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, Jan. 1992.
- [54] A. Aubry, A. De Maio, G. Foglia, C. Hao, and D. Orlando, "Radar detection and range estimation using oversampled data," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 2, pp. 1039–1052, Apr. 2015.

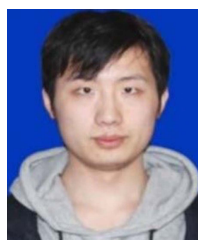
- [55] P. Chen, L. Zheng, X. Wang, H. Li, and L. Wu, "Moving target detection using colocated MIMO radar on multiple distributed moving platforms," *IEEE Trans. Signal Process.*, vol. 65, no. 17, pp. 4670–4683, Sep. 2017.
- [56] H. Li, Z. Wang, J. Liu, and B. Himed, "Moving target detection in distributed MIMO radar on moving platforms," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 8, pp. 1524–1535, Dec. 2015.
- [57] A. Dogandzic and A. Nehorai, "Cramer-Rao bounds for estimating range, velocity, and direction with an active array," *IEEE Trans. Signal Process.*, vol. 49, no. 6, pp. 1122–1137, Jun. 2001.
- [58] S. Kay, *Fundamentals of Statistical Signal Processing: Detection Theory*, vol. 2. Englewood Cliffs, NJ, USA: Prentice-Hall, 1998.



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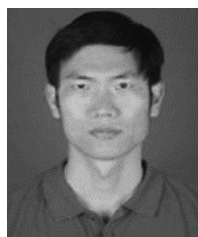
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